

WENDIGO

HUNGER, TABOO, AND THE
CIVILIZATION THAT CANNOT
SAY ENOUGH



A CIVILIZATION THAT EATS WITHOUT LIMIT
DOES NOT FEAST.
IT FEEDS ON ITS OWN FUTURE.

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboaie. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

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*A civilization does not have to want death.
It only has to want one more meal.*

Introduction — A Monster Is a Model

Why Myths Belong in a Technical Argument

This book borrows a name from living Indigenous traditions and therefore begins with a restraint of its own. *Wiindigoo* is an Ojibwe word. The Ojibwe People's Dictionary glosses it as a winter cannibal monster; related figures and spellings occur in several Algonquian-speaking traditions, but they are not one uniform mythology. [1] The creature has been associated, in different tellings, with winter, famine, cannibalism, greed, isolation, and the collapse of obligations that make communal survival possible. [2][3] Modern popular culture has often replaced this morally charged figure with a generic antlered forest demon. That image is largely beside the point.

The argument here is not that an Anishinaabe or Cree teaching can be flattened into a universal theory of economics. It cannot. Nor is this book an attempt to recruit an Indigenous tradition as decoration for a thesis invented elsewhere. The narrower claim is more defensible and more interesting: human cultures repeatedly invented terrifying images for appetite without discipline, and the Windigo is one of the sharpest. Contemporary Indigenous writers, most famously Robin Wall Kimmerer, have already extended the figure toward extractive modern economies and the manufacturing of demand. [4] This book accepts that extension and then moves beyond it. The problem is not capitalism alone. The deeper problem is a civilizational inability to distinguish growth from health, desire from need, possibility from permission, and power from wisdom.

The book will move freely across myths and traditions: Erysichthon in the Greek world, the Buddhist hungry ghost, the language of taboo in Polynesia, the Jewish Sabbath and Jubilee, practices of fasting and restraint, and the modern biological

understanding of cancer. None of these traditions says exactly the same thing. Their differences matter. What unites them here is not a secret perennial doctrine but a recurring human suspicion: **what cannot stop taking will eventually take from the conditions that keep it alive.**

That suspicion deserves to be frightening again.

A modern reader is entitled to distrust a book that moves from monsters to economics, from gods to artificial intelligence, and from cancer to institutions. Analogies can clarify structure, but they can also manufacture false certainty by making unlike things appear equivalent. The discipline of this book is therefore explicit. No myth is offered as scientific evidence. No society is literally a person. No economy has a stomach. No civilization possesses a clinically diagnosable psyche. The myths are models of attention: compact cultural devices that make recurrent forms of danger visible before we have finished naming every mechanism involved.

Science and myth perform different cognitive work. Science decomposes. It asks which variable changes, by how much, under what conditions, and with what uncertainty. Myth compresses. It joins motive, consequence, taboo, memory, and emotion into a scene that can be remembered. A culture needs decomposition to avoid superstition; it also needs compression to avoid a different failure, in which every danger is understood in fragments and therefore never felt as a whole.

The book will repeatedly cross that bridge in both directions. The Windigo image is tested against social dilemmas, adaptive desire, inequality, and institutional selection. Erysichthon is placed beside questions of sacred exclusion, irreversibility, and technological permission. Hungry ghosts are compared cautiously with adaptation and the industrial production of salience. Cancer is used only after its biological mechanisms and the limits of the analogy are made explicit. Thanatos is retained

as a moral endpoint rather than smuggled in as a hidden psychological cause.

This method also prevents the old stories from being treated as decorative props. They are approached as historically situated traditions with meanings that exceed the use made of them here. The title uses the common English spelling *Wendigo*, while discussion of Ojibwe language uses *wiindigoo* and discussion of the broader figure often uses *Windigo*. The variation is not cosmetic. It reminds the reader that the monster did not originate as a global brand. The modern civilizational metaphor is an extension, not an ownership claim.

The Argument in Plain Language

We built civilization by refusing limits. Hunger drove migration. Curiosity crossed oceans. Ambition built cities. Dissatisfaction cured diseases that resignation would have called fate. The capacity to say *not enough* is one of humanity's greatest engines.

But an engine is not a steering system.

At some scale, the same refusal becomes pathological. The organism that cannot stop growing is not called ambitious. It is called cancer. The cell that refuses programmed death is not immortal in any meaningful sense; it participates in a process that can kill the body from which it takes blood. Complex multicellular life exists only because cells accept restraints on proliferation, resource capture, movement, and survival. [12][13][14] A body is possible because its parts do not pursue every local advantage available to them.

Civilization has no comparable guarantee.

The table below prevents a common misunderstanding. The book does not claim that these traditions secretly taught one identical

doctrine. It uses them as different lenses on a recurring systems problem, then tests that problem against modern evidence.

Table I.1 — Five Images, One Architecture

The myths are not equivalent; the table shows only the structural role each plays in this book.

Mythic image	Failure pattern	Modern analytical translation
Windigo	Appetite breaks reciprocity	Externalization, common-pool dilemmas, accumulation without satiety
Erysichthon	A forbidden boundary becomes usable material	Limit illiteracy, irreversibility, possibility mistaken for permission
Hungry ghost	Consumption cannot terminate wanting	Adaptation, positional competition, manufactured salience
Cancer	Local growth escapes organism-level discipline	Breakdown of cooperation, selection for local advantage
Thanatos	Death records what appetite does not count	Residual harm, tail risk, preventable death as externalized cost

Read from left to right, the table moves from image to mechanism. The image supplies memory and moral force; the analytical translation supplies the claims that can actually be debated, evidenced, rejected, or refined.

It has brakes, but many are weakening. Religious prohibitions become embarrassing. Cultural taboos become irrational by definition. Political constraints are redescribed as obstacles to innovation. Ecological boundaries are treated as temporary

engineering challenges. Advertising, recommendation systems, credit, status markets, and increasingly artificial intelligence become technologies for converting latent preference into active demand. Efficiency often reduces the price of consumption and thereby encourages more of it. [18][30] Competition makes restraint strategically dangerous: the state that pauses may lose to the state that does not; the firm that refuses a profitable frontier may be replaced by one that accepts it; the individual who leaves status unclaimed may simply watch someone else take it.

This is how an appetite becomes a system.

The moral horror is not merely that such a system may destroy itself. Long before self-destruction, it can leave other people hungry, untreated, exposed, disposable. According to the 2026 edition of *The State of Food Security and Nutrition in the World*, about 645 million people experienced hunger in 2025. [19] The latest comprehensive UN food-waste estimate, for 2022, found about 1.05 billion tonnes of food wasted at retail, food-service, and household levels. [20] These numbers cannot be naively paired as though discarded food could simply be teleported into every hungry household. Logistics, war, poverty, governance, and access matter. Yet their coexistence destroys a comforting story: global deprivation is not simply the mirror image of absolute global shortage. There are systems capable of producing abundance and systems equally capable of denying access to it.

The same dissonance appears in health. WHO and the World Bank estimated that 4.6 billion people in 2023 were not fully covered by essential health services, while 2.1 billion people in 2022 suffered financial hardship from out-of-pocket health spending. [21] Meanwhile the most technically advanced societies discuss radical longevity, elective enhancement, personalized therapeutics, and machine-accelerated drug

discovery. Again, the point is not that advanced medicine causes the absence of basic care. The point is that a civilization can simultaneously expand the frontier of what is possible and fail to universalize what it already knows how to do.

Wendigo first eats the neighbor. Only later does it eat itself.

Thanatos therefore enters this book not as the cause but as the accountant. Freud's death drive is too speculative to bear the explanatory weight often placed on it, and Freud himself did not use *Thanatos* as the published name of the drive. [10][11] Yet the mythic figure remains useful as an endpoint. A civilization may be animated by appetite rather than by a wish for extinction. Death can still be the cumulative result. The worship of death need not involve black robes, skulls, or an explicit desire to kill. It can consist of treating preventable death as an acceptable externality of keeping the machinery moving.

The argument is therefore deliberately severe:

Civilization's deepest risk may not be hatred of life. It may be an appetite for life, power, growth, safety, status, novelty, and comfort that has forgotten how to stop.

This is not a book against desire. A civilization without desire would be dead already. It is a book against desire that has lost the distinction between *more* and *better*. It is a defense of brakes, boundaries, self-binding, sacred no-go zones, and the ancient idea that some capacities become survivable only when surrounded by prohibitions.

The old world sometimes made too many things taboo. The new world may be making too few.

The thesis can be stated without mythology. Complex systems remain viable only when local units are prevented, by norms, incentives, architecture, or physical constraints, from converting

every available opportunity into private expansion. Human desire is adaptive. Competition can penalize unilateral restraint. Technology can increase the scale and speed with which wants become actions. Many costs are delayed, externalized, distributed, or difficult to reverse. Therefore a civilization whose capabilities grow faster than its capacity for legitimate self-restraint becomes increasingly vulnerable to locally rational actions that aggregate into systemic loss.

Nothing in that argument requires a prediction of inevitable collapse. In fact, recent progress against hunger, the recovery of stratospheric ozone, improvements in many forms of health, and successful environmental regulation show that collective action can reverse harmful trends. The point is not fatalism but asymmetry. Engines of expansion are often rewarded immediately. Brakes are rewarded by events that do not happen. A breakthrough has a photograph; an avoided catastrophe usually does not. The cultural prestige of expansion can therefore grow faster than the prestige of maintenance and refusal.

The moral argument is equally important. Self-destruction is not the first symptom. Long before the system consumes its own substrate, it can consume the margins of weaker people. A civilization may celebrate aggregate abundance while leaving some without food, basic health care, safe housing, political agency, or protection from risks imposed by others. The phrase “eating the future” is dramatic, but the mechanism can be mundane: discounting delayed costs, shifting liabilities, exhausting common resources, or building infrastructures whose benefits are private while failure is socialized.

The proposed remedy is not a return to indiscriminate taboo. Some taboos were instruments of oppression and should remain broken. The book instead argues for *limit literacy*: the capacity to distinguish false barriers from protective boundaries, reversible experiments from irreversible ones, needs from manufactured

appetites, productive competition from escalation traps, and growth that strengthens a system from growth that consumes the conditions of its own continuation. “Enough” is not a doctrine of poverty. It is the name for a stopping rule inside a civilization powerful enough to need one.

Prologue — One More Meal

Imagine a creature that does not hate the forest.

It loves the forest. It loves every tree so intensely that it must possess the timber. It loves the river so completely that it must redirect the water. It loves the mountain for the metal inside it. It loves the animal for the calories in its flesh, the gene in its genome, the molecule in its blood, the price on its skin. It loves the future too: so it borrows from it.

The creature is not a nihilist. It has plans.

It wants warmer houses, longer lives, safer children, richer portfolios, better models, faster processors, larger markets, stronger borders, more resilient armies, more personalized medicine, more convenient travel, more immersive entertainment, more energy, more optionality. It has reasons for each desire. Many of the reasons are good.

This is why the creature is dangerous.

A villain who wants ruin can be opposed. A civilization that wants improvement is harder to restrain. Every new appetite arrives disguised as a local good. Every frontier has beneficiaries. Every limit has a victim who can plausibly say that the limit is unjust to them. Every delay can be framed as cowardice. Every prohibition can be described as irrational fear before progress.

And often it is.

That sentence is important. The history of civilization is full of limits that deserved to be broken: prohibitions on knowledge, caste boundaries, racial exclusions, restrictions on women, inherited hierarchies, blasphemy laws, bans on medical procedures, taboos protecting privilege rather than life. A book that simply praises limits is a book for jailers.

But the opposite superstition is no better: the belief that a boundary is guilty until proven useful.

Some boundaries are the architecture of complex life.

Your body is a federation of cells that could, in principle, compete more aggressively for resources. They do not. Most cells stop dividing when told. They remain in assigned tissues. They accept differentiation. They submit to immune surveillance. Damaged cells can activate programs that end their own existence. The organism is not created by unlimited cellular freedom. It is created by disciplined cellular freedom.

Cancer begins when enough of that discipline fails.

The metaphor is dangerous because it is tempting. Societies are not bodies. Persons are not cells. Political dissent is not a mutation. The history of authoritarianism is filled with rulers who called inconvenient people diseases and then prescribed excision. Nothing in this book licenses that move. The analogy must run in the opposite direction: not from organism to state power, but from multicellular biology to the problem of **cooperation under local incentives**. Cancer biology shows, with brutal clarity, that higher-order complexity requires lower-level restraint. [13][14] It does not tell us who should rule.

The planet now offers an analogous lesson without metaphor. The planetary-boundaries framework reports that seven of nine assessed Earth-system boundaries had been transgressed by the 2025 update, after ocean acidification joined the breached set. [22] The framework is not a divine tablet and the boundaries are not precise cliff edges. They are scientific attempts to identify regions of rising systemic risk. Their existence nevertheless disrupts the infantile version of progress in which every external limit is merely an unopened door.

There are doors behind which there is more room.

There are also doors in airplanes.

The old myths understood the difference by dramatizing it. They placed a monster in the winter. They put a goddess in the sacred grove. They imagined beings whose enormous bellies could never be filled. They surrounded dangerous actions with taboo before anyone could produce a randomized controlled trial proving that violating them was unwise.

Modernity learned to ask a magnificent question: *Why not?*

It has not learned, with equal seriousness, to ask: *What if the prohibition was carrying information we have not yet reconstructed?*

This book is an attempt to rebuild that question without rebuilding superstition.

It begins in winter.

PART I

THE MONSTER

Appetite has always needed a face.



Chapter 1 — The Winter Cannibal

A Monster Born in Scarcity

The winter is long enough that morality acquires a caloric value.

A hunting band can survive bad luck if food is shared. It can survive an injured hunter if someone else carries more weight. It can survive a weak season if stores are distributed with memory of tomorrow. But the margin is thin. One person taking too much is not merely selfish in the abstract. In a closed winter ecology, excess consumption can reappear as somebody else's missing meal.

This is the world in which the Windigo becomes intelligible.

The Ojibwe People's Dictionary gives the stripped-down lexical core: *wiindigoo*, a winter cannibal monster. [1] Scholarship on Windigo narratives repeatedly connects the figure to hunger, greed, winter, cannibalism, and the violation of social obligations. [2][3] One literary analysis of Ojibwe narratives emphasizes the starving season specifically: in a setting where food is scarce, taking more than one's share can amount, morally, to eating into the bodies of others. [2] The monster is therefore not only a predator. It is a human appetite that has crossed a line after which the community can no longer recognize itself in the eater.

That line matters.

The first discipline required by this book is to resist the modern reflex to turn the Windigo into a mascot. In living Anishinaabe traditions, the figure belongs to particular languages, communities, histories, and teachings. The Ojibwe People's Dictionary gives the lexical core with admirable economy: *wiindigoo*, “a winter cannibal monster.” That definition matters because winter is not scenery. Winter changes the ethics

of appetite. When stored food is finite and the next successful hunt is uncertain, taking too much is not merely a private vice. It changes the survival probabilities of everyone around the fire. The monster therefore condenses a social truth into a body that can be feared: appetite becomes monstrous when it severs the eater from reciprocity.

Modern readers often encounter a horned creature in a dark forest. That image is visually powerful but conceptually weak. The antlers are not the argument. The argument is that a person can become unable to recognize other persons as members of the same survival system. Cannibalism is the most violent possible image of that failure because it eliminates the distance between resource and neighbor. The consumer no longer destroys a remote landscape or exploits an invisible worker. The resource has a face. The meal is somebody whose continued existence should have limited the eater's claim.

This is also why the metaphor must not be reduced to a sermon about greed. Greed is a trait attributed to individuals; the civilizational problem is a pattern produced by institutions, technologies, and competition. A person may be moderate while working inside a firm that must grow, a state that must accumulate strategic capacity, or a platform that must increase engagement. The old story asks whether an individual has become a monster. The modern question is harsher: can a system select for monstrous outcomes even when most participants do not feel monstrous at all?

That shift from character to structure will guide the rest of the book. The myth gives us a face. Systems analysis asks how the face can emerge from ordinary incentives. The combination is useful because technical language can explain a feedback loop without making us feel its moral temperature, while myth can make us feel danger without telling us how it is produced. We need both.

Cannibalism is uniquely useful as a moral image because it collapses the usual distance between consumption and victim. In ordinary economic life, the causal chain is long. A cheap object has a worker somewhere. A full tank has a landscape somewhere. A rare mineral has a mine somewhere. A portfolio has externalities somewhere. A comfortable institution has a queue outside it. Modern systems are very good at lengthening chains until the end of a transaction disappears from perception.

Cannibalism shortens the chain to zero. The eater sees the eaten.

This is one reason the image has survived modernity. Wendigo is what appetite looks like when the externality is placed back on the plate.

Contemporary Indigenous writers have used the figure precisely because modern economies can reward what older stories condemned. Kimmerer's language of a Windigo economy is a critique of fabricated demand, overconsumption, and extraction without reciprocity. [4] Other scholarship traces Windigo imagery through environmental injustice, colonial extraction, and the conversion of living landscapes into inventories. [3] The extension is powerful, but it risks becoming too easy if Windigo is simply another name for a political enemy.

A monster that always lives in the other party's house teaches nothing. The deeper hypothesis is that Windigo is a recurrent failure mode of organized intelligence.

An imperial court can become Windigo. A corporation can. A revolutionary state can. A military bureaucracy can. A university can. A household can. A research field can. A digital platform can. A person can. The common pattern is not ownership structure but **appetite coupled to positive feedback and insulated from the cost of appetite.**

From Winter Camp to Growth Economy

Success provides the capacity for more success. Capacity enlarges ambition. Ambition justifies infrastructure. Infrastructure creates constituencies that depend on continued expansion. Expansion normalizes the new scale. The new scale becomes the baseline from which any pause feels like decline.

The creature grows because it eats. It becomes hungrier because it has grown.

This loop is visible in ordinary life. A larger home creates more surfaces to furnish and maintain. Higher status changes the reference group against which status is measured. More compute makes previously impossible models feasible, which makes previously absurd compute budgets appear prudent. Greater military capability reveals new vulnerabilities that justify greater military capability. Efficiency reduces the effective price of a service and can produce rebound: the service is used more because each unit is cheaper. [18][30]

None of these mechanisms requires madness. That is the central discomfort. Windigo can be produced by rational agents following local incentives.

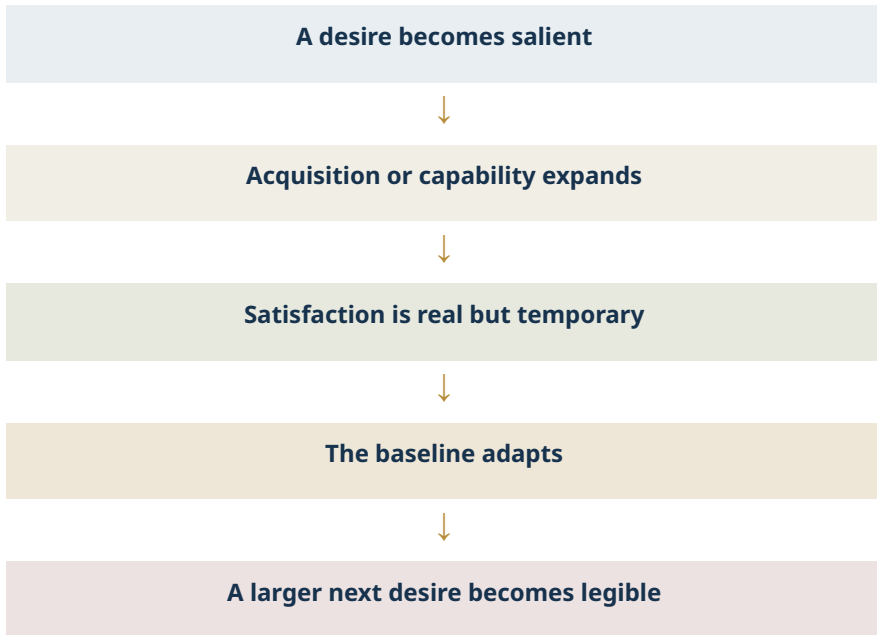
The winter camp and the global economy are not the same system, but they share one feature that matters: neither can repeal dependence. A modern supply chain can move grain across oceans, desalinate water, synthesize fertilizer, discover new energy sources, and substitute one material for another. These capacities make the system vastly more flexible than a small community facing a failed hunt. Yet flexibility is not independence. Every economy still sits inside material, ecological, biological, and social substrates that it did not create from nothing.

The Windigo dynamic is easiest to understand as a feedback loop rather than as a personality defect. The diagram deliberately

contains no moral label; every step can occur in ordinary, healthy motivation.

Figure 1.1 — The Appetite Loop

How successful satisfaction can create the conditions for renewed demand.



The dangerous transition occurs when institutions reward only the forward motion of the loop and never represent satiety as a valid terminal state. The loop then ceases to be merely psychological and becomes economic and political infrastructure.

The danger begins when flexibility is mistaken for exemption. Because one limit was moved yesterday, today's limit is interpreted as merely another engineering inconvenience. Because scarcity in one domain was overcome, scarcity as a category begins to look intellectually obsolete. This is one of modernity's most productive habits and one of its most dangerous. The same mentality that refuses premature resignation can become unable to distinguish a movable constraint from a load-bearing boundary.

Growth also changes the selection environment. When actors compete, voluntary restraint can be punished even when restraint would be collectively beneficial. A company that leaves a profitable resource untouched may lose market share. A government that slows a strategically important technology may fear dependence on a rival that continues. A media platform that declines manipulative engagement techniques may watch attention migrate elsewhere. Under such conditions, appetite does not need to be celebrated morally. It only needs to survive competitively better than restraint.

The result is a civilizational version of what the winter story dramatizes at human scale. The group still depends on a shared substrate, but the rules increasingly reward the participant who can appropriate more of it, faster, while moving the consequences outside the accounting frame. This is why the Windigo image remains useful in a world of satellites and machine learning. The monster is not an argument against complexity. It is a warning about complexity that has become skilled at hiding the mouth from the body it feeds upon.

The historical change in scale also changes visibility. A small community can often see who has eaten, who is hungry, and what remains in storage. A global system sees through indicators, contracts, prices, models, and administrative categories. That abstraction is indispensable, but it can separate action from consequence. The person who benefits from a cheap product may never see the mine, warehouse, algorithmic labor market, river, or family through which its price became possible. The modern moral problem is therefore not merely excess; it is excess combined with distance. The Windigo grows more plausible when the eater can sincerely believe there is no one at the other end of the bite.

Imagine two firms. One imposes a voluntary ceiling on extraction because its leaders judge further expansion

ecologically reckless. The other does not. If markets reward the second firm, capital migrates. Imagine two states. One decides that a class of strategic AI is too dangerous to develop. The other believes development is essential for security. If the second gains decisive advantage, the restraint of the first may look retrospectively like negligence. Imagine two individuals in a status market. One opts out. The other accumulates. If accumulated status buys influence over the rules, opting out does not neutralize the game; it may transfer control to the hungrier participant.

The problem is not simply desire. It is selection among desires. Systems can preferentially promote actors that tolerate fewer limits.

This is why sermons about personal moderation are insufficient. A culture can admire restraint while its institutions eliminate the restrained. A CEO who says "enough" inside a system that prices only growth may be replaced. A politician who refuses the next round of competitive escalation may lose. An academic who publishes only when something matters may disappear from metrics constructed around volume. A user who refuses engagement optimization is replaced, in platform statistics, by the one who keeps scrolling.

Windigo is therefore not a personality diagnosis. It is a selection environment. The old story frightened individuals so that the community could survive winter. The modern version must frighten systems. And the most frightening feature is not that the monster is outside the camp. It is that the camp may have redesigned itself to reward whoever eats fastest.

The first sign of a Wendigo civilization is not hunger. It is the inability to recognize satiety as an achievement.

Chapter 2 — Erysichthon's Axe

The Tree Outside Calculation

Before the first bite comes the first prohibition.

In the Greek story, Erysichthon enters a grove sacred to Demeter and cuts down a tree he has been warned not to cut. The details vary across ancient tellings, but the structure is stable enough to survive two millennia: sacrilege is followed by insatiable hunger; appetite consumes wealth, household, and finally, in Ovid's version, the body of the eater himself. [5][6][7]

It is difficult to design a cleaner myth for modernity.

The grove is not merely a stock of timber. It is a place removed from ordinary calculation. Its sacredness means that the usual question — *what can this be used for?* — has already been answered: not everything available is available for use.

Erysichthon refuses the category.

The axe is an epistemology. It says that a tree is a tree, wood is wood, and invisible restrictions imposed by priests, stories, gods, or ancestors have no standing against material utility. From one perspective, he is the modern hero before his time. He demystifies. He converts sacred capital into usable resource. He declines to let inherited narrative veto present action.

Erysichthon's story adds something that the winter monster cannot. The decisive event occurs before hunger: a boundary is declared, explained in the language of sacredness, and deliberately crossed. In Callimachus, the grove of Demeter is dense, watered, and beloved by the goddess. In Ovid, the sacred tree carries votive signs. The point is not that ancient people possessed a modern theory of ecosystem services. The point is that the grove has been removed from ordinary appetite by a rule stronger than price.

A sacred prohibition is epistemically strange to a modern mind. It does not say that cutting the tree is inefficient at today's prices. It does not say that an expert panel has calculated a negative expected value. It says, before calculation begins, that this thing is not available for this use. Such categorical rules can be foolish, oppressive, or corrupt, and history contains more than enough examples. Yet the form of the rule solves a problem that optimization often cannot: it prevents every value from being converted into a tradable quantity inside the same decision procedure.

The distinction matters whenever benefits are immediate and concentrated while costs are delayed, dispersed, uncertain, or impossible to restore. A sacred grove may be scientifically misunderstood and still be institutionally protected. A fully priced forest, by contrast, remains vulnerable to a sufficiently high price. The argument is not that sacredness is superior to science. It is that a civilization needs some mechanisms capable of saying "not for sale" or "not an admissible move" when the information required for a perfect calculation does not exist.

Erysichthon is therefore a useful character for an age that admires optionality. He cannot tolerate an excluded option. The tree's refusal to become timber appears to him as irrational obstruction. His tragedy begins not with excessive consumption but with the inability to recognize that some of the world's value may depend precisely on remaining outside his plan.

Then the hunger comes. The punishment is not lightning. It is appetite.

That choice is extraordinary. Demeter does not simply kill the violator. She turns his own desire into the instrument of destruction. The more he consumes, the more urgent consumption becomes. Wealth is liquidated into food. Relations become instruments. The household becomes fuel. At the end, the distinction between eater and resource collapses.

The story is not a primitive environmental regulation hiding in theatrical costume. It is more ambiguous than that. Sacred groves could serve religious, political, ecological, and social functions simultaneously. Ancient myths should not be retrofitted into contemporary policy papers. Yet Erysichthon stages a problem that remains exact: **when every prohibition must justify itself inside the very calculus it was designed to restrain, the calculus has already won.**

Suppose a forest is forbidden because it is sacred. The utilitarian asks for measurable benefits. Carbon storage? Biodiversity? Watershed stability? Tourism? Mental health? Cultural heritage? Good. Those answers may save the forest.

But notice what has happened. The forest survives only by submitting to valuation.

If a future technology replaces its carbon storage, engineers its species elsewhere, filters its water, simulates its beauty, and compensates its cultural loss, does the prohibition disappear? In a purely instrumental framework, perhaps it should. The sacred has been translated into a bundle of services, and bundles can be substituted.

A taboo says something structurally different: **this is not in the optimization domain.**

That sentence is dangerous. It has protected tyrannies as well as forests. Human beings have declared rulers sacred, castes untouchable, women impure, texts unchallengeable, and minorities forbidden. The sacred can become a shield for domination. The proper conclusion is not that sacred boundaries are always wise. It is that a civilization needs a category of constraints strong enough to survive short-term optimization, while also needing institutions capable of revising those constraints when they protect cruelty rather than life.

When Possibility Becomes Permission

Modernity became brilliant at the second task and suspicious of the first.

We inherited the axe and forgot the grove.

The result is not that everything has been destroyed. It is subtler. Everything must now appear before a tribunal of use.

A species is useful because it supports ecosystem services. A human relationship is useful because it improves well-being. Sleep is useful because it raises productivity. Parks are useful because they increase property values. Arts are useful because they improve cognition. Privacy is useful because trust helps markets. Democracy is useful because it produces stability. Even rest must present a business case.

The language sounds rational because it is rational. It is also ravenous.

A thing that cannot defend itself in the vocabulary of utility becomes vulnerable to whatever can. The sacred grove, by contrast, begins from a refusal: there are values whose protection cannot depend entirely on proving that they produce more of some other value.

Modern societies are rightly proud of having broken prohibitions that protected ignorance, hierarchy, and cruelty. Medicine advanced by refusing taboos around anatomy and disease. Science advanced by allowing questions that religious or political authorities had forbidden. Liberal institutions widened the lives of people whose possibilities had been constrained by caste, sex, race, or inherited status. Any defense of limits that forgets this history becomes a defense of cages.

But liberation can generate its own superstition: the belief that a boundary is presumptively illegitimate because some boundaries were illegitimate. In that frame, the burden of proof

falls almost entirely on restraint. The proposed action arrives with visible beneficiaries and measurable gains; the prohibition must defend losses that may be probabilistic, long-term, distributed across generations, or embedded in relationships that resist quantification. The axe speaks in the language of opportunity. The grove must learn to speak finance, law, risk, and metrics merely to remain standing.

Technological competition intensifies this asymmetry. Once a capability exists in principle, actors begin to ask who will develop it first, who will control it, and what will happen if rivals move while they wait. Possibility becomes strategic pressure. Strategic pressure becomes moral language: delay is irresponsibility, restraint is surrender, caution is betrayal of the patients, citizens, shareholders, or nation that might benefit. The original question—whether this capability should be pursued at all—quietly disappears.

This does not mean that civilization should freeze discovery. It means discovery requires a second intelligence alongside inventive intelligence: the capacity to preserve the legitimacy of refusal. A mature technological culture must be able to say that some experiments should be delayed, some markets should remain uncreated, some data should remain uncollected, some ecological assets should remain unconverted, and some efficiencies are not worth the fragility they purchase. Erysichthon becomes modern when the axe is no longer held by one arrogant king but distributed across institutions that each possess a reasonable argument for another cut.

The distinction between possibility and permission is already familiar in other domains. Medical research can technically perform experiments that ethics committees will not authorize. Courts can sometimes obtain useful evidence by means that constitutional orders refuse to permit. Engineers can build systems whose operating envelope deliberately excludes

conditions that would produce more performance. In such cases, restraint is not treated as hostility to knowledge or capability. It is part of what makes the capability socially admissible. The strange feature of contemporary technological culture is not that it lacks limits entirely, but that it often treats limits as temporary embarrassments rather than as constitutive parts of responsible power.

Erysichthon's error is therefore not merely greed. It is **limit illiteracy**. He cannot read a prohibition except as an obstacle. Civilizations can suffer the same defect.

The Manhattan Project proved that a physical possibility could become a geopolitical imperative with terrifying speed. Contemporary biotechnology, artificial intelligence, geoengineering, and autonomous weapons all reopen the same question in different forms: once a capability exists in principle, what social mechanism can credibly keep it unused? "Do not" is weak against "someone else will." Ethical reluctance is weak against strategic competition. Regulation is weak against jurisdictional arbitrage. Public fear is weak against private reward.

The axe has become distributed. No single Erysichthon need enter the grove. Thousands of reasonable actors can each cut a branch.

At the end of the myth, hunger is no longer an appetite for external goods. It is an algorithm consuming its own substrate.

Civilizations can do this too. They can exhaust soil to preserve agricultural output, erode trust to win political cycles, spend future fiscal capacity to protect present consumption, destroy attention to monetize information, consume institutional legitimacy to achieve short-term victories, and burn ecological resilience to keep energy cheap.

Each bite solves a problem. Then the problem returns larger.

The old myth does not tell us never to cut trees. It tells us to fear the person — or system — that can no longer imagine a tree it is forbidden to cut.

Chapter 3 — The Hungry Ghost

A Belly Larger Than Its Throat

The hungry ghost has an enormous belly and a throat that cannot satisfy it.

Across Buddhist traditions, *preta* came to denote beings tormented by deprivation, especially hunger and thirst; in later iconography they are often represented with swollen bellies and constricted throats. [8] Buddhist accounts connect the state with greed, miserliness, attachment, and craving. The philosophical machinery behind the image is richer than a moral fable about gluttony. The relevant concept, *taṇhā*, literally carries the sense of thirst: craving that grasps at pleasurable states, resists painful ones, and keeps rebuilding dissatisfaction around what is absent. [9]

The Windigo eats others.

Erysichthon eats the world and then himself. The hungry ghost cannot even successfully eat. Together they map three stages of the same civilizational pathology.

The hungry ghost changes the geometry of appetite. The Windigo is terrifying because it consumes others. Erysichthon is terrifying because appetite turns back upon the eater. The *preta* is terrifying because consumption itself has become incapable of completing the act of satisfaction. In many Buddhist traditions, hungry ghosts are imagined with swollen bellies and narrow throats, tormented by hunger and thirst; greed and miserliness are among the moral causes associated with this realm. The image is not a clinical model of addiction, but it captures a distinction modern psychology recognizes in another vocabulary: wanting and liking are not the same process, and satisfaction is not guaranteed to scale with acquisition.

This distinction becomes civilizationaly important once basic deprivation has been reduced. A hungry child needs food, not a philosophy of detachment. A patient without antibiotics needs treatment, not an argument about hedonic adaptation. The ethics of “enough” begins with sufficiency, not with romanticizing scarcity. Yet after sufficiency, human evaluation remains relative, adaptive, and socially situated. Aspirations rise. Reference groups change. Luxuries become baselines. Improvements that once transformed life can become invisible background conditions.

Research on aspirations and subjective well-being provides a modest empirical anchor for this intuition. Studies such as Knight and Gunatilaka's work in rural China found that higher income could raise well-being while rising aspirations simultaneously eroded part of that gain. [16] The result is not the simplistic claim that money does not matter. Money matters enormously, especially when it buys security, health, time, and dignity. The more careful claim is that the psychological mechanism that registers “enough” does not behave like a fixed tank with a visible fill line.

A civilization that can expand supply but cannot cultivate satiety may therefore solve material scarcity while preserving motivational scarcity. It can make the belly larger while leaving the throat narrow. That is the hungry ghost's warning in modern form.

The ghost also exposes a mistake in the rhetoric of abundance. Abundance is morally important when it removes coercive scarcity. A person who no longer fears hunger, untreated illness, or homelessness gains real freedom. Yet the psychological experience of abundance depends partly on attention. If the field of comparison expands faster than material conditions improve, a person can inhabit objective plenty while experiencing subjective insufficiency. This does not make the insufficiency imaginary; social comparison can affect real opportunities. It

means that production policy alone cannot answer the question of what a prosperous civilization is for. A society that knows how to multiply goods but not how to cultivate attention, gratitude, purpose, and non-positional sources of dignity may become materially rich while remaining motivationally poor.

First, desire violates reciprocity. Then it consumes the conditions of its own satisfaction. Finally, satisfaction itself becomes structurally impossible.

This third stage deserves more attention because modern abundance is often imagined as the cure for the first two. If hunger drives conflict, produce more. If scarcity drives competition, automate more. If people fight over energy, make energy abundant. If housing is positional, build. If intelligence is scarce, manufacture it. If labor is expensive, mechanize it. If life is short, extend it. The formula is seductive because it has worked repeatedly.

For many forms of suffering, more really is better.

A child with malaria does not need a lecture on spiritual craving. The child needs diagnosis and medicine. A household without food needs calories before philosophy. A city without sanitation needs pipes. Romanticizing deprivation is one of the uglier luxuries of the comfortable.

But after sufficiency, another mechanism becomes visible.

Aspirations adapt. Reference groups move. New capabilities generate new baselines. Research on income aspirations and subjective well-being has repeatedly found that people's standards can rise with their own income and the income of those around them. [16] Social comparison means that some goods function less as absolute benefits than as positions in a ranking. A larger house can improve life; it can also redefine what counts as a normal house. More computing power solves problems; it also makes previously extravagant applications

ordinary. Faster networks remove waiting; they also create services that assume permanent availability.

The throat narrows as the belly grows.

Civilization is especially vulnerable because it can industrialize aspiration. The hungry ghost of old was trapped by karma. The modern hungry ghost has a marketing department.

An economy can discover needs. It can also manufacture salience. It can turn the absence of a product into a felt deficiency. A platform can observe which stimuli retain attention and amplify them. A recommendation model need not know what a good life is; it needs only a gradient. A financial system can convert future income into present purchasing power, allowing today's desire to recruit tomorrow's labor. A prestige economy can transform universal access into a reason to seek rarer forms of distinction.

The Industrialization of Dissatisfaction

The problem is not conspiracy.

No committee needs to decide that humanity should remain dissatisfied. Dissatisfaction is commercially legible. Satiety is not.

A customer who says "I have enough" is, from the perspective of many business models, a disappearing market.

This creates a profound asymmetry. Technologies of appetite are scalable. Technologies of satiety are not.

There are industries for food, entertainment, fashion, travel, gaming, pornography, luxury, pharmaceuticals, fitness, education, self-improvement, political outrage, social connection, spirituality, and increasingly simulated companionship. There is no comparably powerful sector whose revenue depends on people concluding, with durable confidence, that they need nothing more from the sector.

Even wellness can become another appetite.

The hungry ghost is useful here because it shifts the critique from objects to the **architecture of wanting**. The issue is not whether a particular desire is vulgar. A civilization can become insatiable for noble things: safety, knowledge, health, justice, longevity. There is no obvious quantity at which any of these becomes complete.

How safe is safe enough? How much knowledge is enough?

The decisive modern innovation is not merely that we can satisfy more wants. It is that we can detect, stimulate, personalize, and continuously refresh them. Advertising began this process long before digital platforms. Credit separated desire from current purchasing power. Fashion accelerated the social obsolescence of otherwise functional goods. Recommendation systems now observe behavior at enormous scale and learn which sequence of stimuli keeps attention moving.

None of this requires a conspiracy to make humanity miserable. The mechanism is more ordinary. In many markets, a satisfied customer is economically less active than a customer whose next desire can be made salient. A platform whose user closes the application because life feels complete has lost engagement. A service that perfectly cures a recurring need may be less profitable than one that converts the need into a subscription. Firms are not uniquely wicked for responding to incentives; the point is that the incentives can create a civilization in which the production of dissatisfaction becomes technically sophisticated.

Artificial intelligence may deepen this asymmetry. Systems that model individuals can learn not only what people state that they want but when they are tired, lonely, impulsive, status-sensitive, frightened, or susceptible to social proof. The same predictive capacity can be used to help a person meet chosen goals or to keep the wanting system activated. The

difference is not in the mathematical sophistication. It is in the objective being optimized and the constraints surrounding it.

This is why the hungry ghost belongs in a book about civilization rather than only in a book about consumer culture. Artificial hunger can attach itself to safety, certainty, longevity, political purity, technological capability, visibility, and moral approval just as easily as to shoes or screens. Any value can become ravenous when it has no stopping condition. The civilizational question is therefore not whether desire can be eliminated. It is whether institutions can recognize the moment when serving a desire turns into maintaining the machinery that produces it.

The most revealing business model is one that would be threatened by the permanent success of its own stated mission. If a wellness service profits only while users remain dissatisfied with themselves, or an attention platform only while people remain unable to leave, the system contains a conflict between welfare and revenue. This does not prove exploitation in every instance, but it gives regulators, designers, and citizens a diagnostic question: what happens to the institution if the user becomes genuinely satisfied? A civilization capable of asking that question begins to distinguish industries that profit from solving problems from industries that profit from preserving the conditions under which the problem remains commercially active.

How long should a life be? How much surveillance is justified to prevent one more attack? How much computation is justified to gain one more point of capability? How much intervention is justified to eliminate one more source of discomfort? Once improvement is framed as an unbounded moral duty, refusal becomes morally suspicious.

This is where the hungry ghost meets the technocrat. The most dangerous appetite may be the appetite that believes itself to be compassion.

A society committed to eliminating every risk can destroy privacy, spontaneity, and autonomy. A society committed to maximizing health can begin to treat ordinary variance as pathology. A research culture committed to knowledge can generate mountains of trivial work because the metric rewards production rather than understanding. A political culture committed to justice can expand categories of control until every interaction becomes administrable.

The ghost does not need luxury. It needs a value that has no stopping rule.

Any good without a stopping rule can become an appetite. Any appetite embedded in competition can become a machine.

The Buddhist answer is not annihilation of all desire. Even within Buddhist traditions, wholesome aspiration is distinguished from craving. [9] The distinction matters for civilization too. Desire can orient action toward growth, care, learning, and creation. Craving is desire that reproduces itself by consuming its own satisfaction. The task is not to become empty. It is to recover a throat large enough to receive what the belly already has.

Chapter 4 — Thanatos Keeps the Ledger

Death as an Outcome, Not a Desire

Death is patient. Appetite does not need to worship it.

The Greek Thanatos is not the Windigo. He does not gorge. He is death personified: a boundary, an ending, a power no appetite can bargain with forever. Later psychoanalytic language attached his name to Freud's controversial death drive, though the term *Thanatos* was not Freud's own published label. [10] Freud's underlying question, however, was unmistakably civilizational. In *Civilization and Its Discontents*, he asked whether cultural development could master human aggression and self-destruction at the very moment technology had made species-level extermination imaginable. [11]

The twentieth century made that question impossible to dismiss.

The twenty-first makes it stranger.

Thanatos enters this book as a corrective to a tempting mistake. It is intellectually easy to look at recurrent war, ecological damage, nuclear arsenals, engineered risks, and self-defeating institutions and say that civilization must secretly desire death. That move is rhetorically dramatic and empirically weak. Freud's death drive was itself speculative, and the popular label *Thanatos* was not his standard published term. A civilization does not need an unconscious wish for extinction to behave in ways that make extinction more likely.

The distinction between motive and outcome is essential. A pilot can crash while trying to arrive faster. A company can destroy a market while trying to dominate it. A state can make itself less secure through an arms race undertaken in the name of

security. An agricultural system can reduce long-term fertility while maximizing short-term output. The causal engine in each case is not love of destruction but optimization inside an incomplete frame.

Freud nevertheless posed a question that remains uncannily modern. In *Civilization and Its Discontents*, he asked whether cultural development could master aggression and self-destruction at a time when humans had acquired enough control over natural forces to exterminate one another. Nuclear weapons later made the sentence literal. Today the portfolio of high-consequence capabilities is broader: advanced biotechnology, cyber-physical infrastructure, autonomous systems, and increasingly capable AI can concentrate or accelerate forms of agency that were once slow, local, or expensive.

Thanatos therefore functions here as the name for a destination that does not need to be chosen. Appetite, competition, fear, and technological power can walk toward death while carrying banners that say prosperity, security, health, and progress. The danger is precisely that the route can remain morally legible to its travelers almost until the end.

This framing also protects the argument from melodrama. Nuclear war is not frightening because leaders must be secretly suicidal; it is frightening because deterrence, misperception, signaling, command systems, and compressed decision time can interact in ways no participant desires. Environmental overshoot is not frightening because consumers hate nature; it is frightening because millions of ordinary benefits can accumulate into system-level pressure. The death ledger is most important exactly where intention is least spectacular. If catastrophic outcomes required theatrical villains, civilization would be safer than it is.

Our most plausible routes to catastrophe are not all powered by a desire to die. Nuclear escalation can emerge from deterrence doctrines intended to prevent war. Ecological overshoot can emerge from billions of ordinary acts intended to improve lives. Artificial intelligence risk, where one judges it significant, can emerge from competition to build useful systems. Antibiotic resistance can grow from attempts to cure disease. Financial crises can be created by instruments designed to allocate risk. Information disorder can be amplified by platforms designed to connect people.

Death can be the residual of optimization.

This is the role Thanatos plays in the book. He is not the secret deity humanity consciously serves. He is the endpoint that records what appetite refuses to count.

Every system has a ledger. Some columns are visible, others externalized.

A factory counts output and cost. It may not fully count respiratory disease outside the fence. A food system counts yield and price. It may not fully count soil loss, antibiotic resistance, or the metabolic consequences of diets designed around cheap calories. A military system counts deterrence credibility. It may not be able to price the tail risk of accidental escalation. A platform counts engagement. It may not know how to value degraded attention or social trust. A national economy counts production. It does not subtract every depleted aquifer, broken community, or lost species as if these were ordinary depreciation.

Thanatos lives in the unpriced column.

This is not an argument that everything important can or should be priced. The opposite. The fantasy of complete accounting is itself a form of appetite: the desire to convert moral judgment into a dashboard. Some losses are not commensurable. The death of a child cannot be offset, in any humanly serious

sense, by enough gains elsewhere. A species cannot be restored by moving digits between accounts after extinction. A community's trust, once destroyed, may not return because a cost-benefit model later recognizes the error.

The Unpriced Column

The ledger metaphor is therefore accusatory, not technocratic.

It asks what a civilization is willing to let disappear while still calling itself successful.

The modern cult of death does not require devotees chanting for extinction. Its priests may be competent, decent people presenting slides about acceptable losses.

This sounds unfair until one notices how much governance is precisely the distribution of acceptable loss. Road systems trade mobility against deaths. Health systems allocate finite resources. Environmental rules trade production against exposure. Military doctrines explicitly model casualties. Every serious society does this because zero risk is impossible.

The moral problem is not that trade-offs exist.

It is that trade-offs become invisible when the people receiving benefits and the people carrying costs are separated by geography, class, generation, or abstraction.

One person's cheap food can be another person's polluted water. One generation's debt can be another's constraint. One state's security can be another population's permanent fear. One industry's growth can become a community's disease burden. One user's convenience can become millions of low-paid workers hidden inside the service chain.

Every institutional system contains a boundary around what it sees. Budgets record some costs and omit others. Performance indicators celebrate some outcomes and make other losses disappear. A factory can measure units produced, energy

consumed, labor hours, and financial margin with exquisite precision while respiratory disease in a distant community remains statistically and morally remote. A platform can optimize clicks while loneliness, polarization, or degraded attention appear only later in academic studies. A state can measure deterrence capability while the probability of accidental escalation remains a model rather than a lived event.

Economists call many such effects externalities, but the word can sound cleaner than the phenomenon. An externality is often simply a cost that the decision-maker has succeeded in placing outside the decision. Some can be priced. Others involve irreversible ecological change, loss of cultural meaning, dignity, non-consenting future persons, or tail risks whose probabilities are uncertain and whose consequences cannot be compensated after the fact. The fact that a value resists pricing does not make it unreal; sometimes it means the value is too structurally important to be treated as another line item.

Thanatos keeps the ledger of what disappears from the optimizing frame. Preventable deaths, species lost, soils depleted, trust exhausted, children whose opportunities are narrowed, and future options foreclosed can all be real even when no actor intended them. The ethical accusation is not that every loss proves malice. It is that repeated invisibility becomes a form of permission.

A limit-aware civilization would therefore ask a different question from “did the decision-maker intend harm?” It would ask whether the system is organized so that some classes of harm are predictably shifted onto people with less power, onto nonhuman life, or onto the future. The moral seriousness of a system is visible not only in what it rewards but in which losses it allows to remain off-book.

A useful test is to ask who is absent when success is declared. Is the affected community in the room? Are children

represented? Are future maintenance costs included? Are ecological effects measured beyond the project boundary? Are rare but catastrophic failures treated as more than inconvenient tails of a distribution? No decision procedure can include everything, and attempting to do so would paralyze action. The point is not total accounting. It is institutional humility: the recognition that a clean balance sheet may indicate a narrow boundary rather than a clean reality.

The further the victim from the decision, the easier appetite becomes. This is where Wendigo and Thanatos finally meet. Wendigo says: more. Thanatos says nothing. The silence is the point. Death does not negotiate with the narrative that produced it.

A depleted aquifer does not care that extraction was profitable. A collapsed fishery does not care that every boat acted rationally. A destabilized climate does not care that no individual flight caused it. A person who could not afford a medicine does not become less dead because the system that priced it was internally coherent.

Civilization's moral maturity can therefore be measured partly by what it refuses to externalize into silence.

The frightened reader may now expect a familiar solution: better regulation, better taxation, better technology, better ethics. All are useful. None reaches the root. The root is appetite under conditions where consequences are distant and limits are negotiable. To confront that, we need to understand something older than policy. We need to understand why human beings invented forbidden lines.

PART II

THE BODY

Complex life begins where appetite accepts discipline.



Chapter 5 — The Animal That Cannot Say Enough

Hunger, Reward, and Adaptation

Human beings are not uniquely greedy. They may be uniquely capable of turning greed into infrastructure.

A squirrel caches food. A wolf defends territory. A chimpanzee competes for rank. Biological organisms evolved under scarcity, uncertainty, predation, sexual selection, and status competition long before anyone invented a derivative contract or a shopping mall. Appetite is not a cultural error pasted onto an otherwise serene animal. It is part of the machinery that kept ancestors alive.

This makes moralizing about desire intellectually cheap.

A population that felt no hunger would starve. A primate indifferent to status might fail to reproduce. A species without curiosity would not explore. A child satisfied with ignorance would not learn language. A scientist satisfied with the current theory would not discover its failure. The organism that never pushes against a boundary can die behind a boundary that could have been moved.

The problem begins when a mechanism shaped for local scarcity acquires planetary reach.

Evolution did not design appetite for supermarkets, algorithmic persuasion, consumer credit, fossil energy, global logistics, synthetic biology, or machine intelligence. It did not need to solve the problem of a primate whose individual choices could participate in altering atmospheric chemistry. It did not anticipate a status competition visible to billions of observers or a marketplace capable of delivering a stimulus within seconds of its desire becoming conscious.

Human appetite is not a single mechanism. Biological hunger, curiosity, sexual desire, attachment, status seeking, novelty seeking, fear reduction, and the search for meaning involve different neural and cultural systems. It would be pseudo-scientific to collapse them into one “Wendigo circuit.” The book’s claim is more abstract: many motivational systems are adaptive rather than self-terminating. The baseline against which improvement is judged can move.

This is ordinarily useful. An organism that remained permanently euphoric after finding one meal would not prepare for the next winter. A scientist who felt final satisfaction after one discovery would stop learning. A society that treated the first improvement in public health as sufficient would leave preventable suffering untouched. Adaptation keeps behavior responsive to changing conditions.

The problem appears when a mechanism evolved or learned for local adaptation is inserted into institutions that can scale its objects indefinitely. Status is a clear example. Status is relational. Not everyone can occupy the top decile at once. If material consumption becomes a primary language for displaying position, gains in absolute abundance do not abolish competition; they change the symbols through which competition is expressed. The larger house, rarer experience, better school district, more exclusive network, more visible body, more capable model, or more prestigious title becomes meaningful partly because others do not have it.

This helps explain why a society can become richer without becoming proportionally easier to satisfy. The problem is not that well-being never improves. It does. The problem is that some forms of progress generate new reference points and new arenas of comparison. An appetite that was once bounded by physical scarcity can become bounded only by the existence of somebody else with more.

We are old appetites with new actuators.

The mismatch appears in several well-studied forms. People adapt to changed circumstances. Aspirations can rise with income. Reference groups change as status changes. [16] The satisfaction delivered by an improvement can be real while still becoming normalized. This is not evidence that wealth is useless. Escaping poverty has enormous consequences for health, agency, education, safety, and dignity. But it does mean that the subjective word *enough* does not automatically appear when material conditions cross an objective threshold.

A civilization that waits for spontaneous satiety may wait forever.

Status makes the problem worse because many status goods are inherently relative. If everyone receives the same improvement, the absolute benefit may remain while the ranking benefit disappears. The point of a larger office, more selective credential, rarer address, more exclusive experience, or more visible achievement can partly be that others do not have it. Such competitions cannot be solved simply by producing enough of the scarce thing, because scarcity is part of the product.

Then there is the Red Queen.

Evolutionary and organizational research uses the Red Queen metaphor for competitions in which actors must keep changing simply to maintain relative position. [17] The logic is easy to recognize in arms races, business innovation, academic prestige, cybersecurity, and technological competition. One actor improves; the improvement changes the environment for everyone else; rivals improve; the baseline moves. Nobody receives a final victory because success rewrites the conditions of success.

This is appetite without indulgence. The runner may hate running. It must run anyway.

That distinction matters because it breaks the comforting idea that the Wendigo problem can be solved by teaching individuals to want less. A state may prefer restraint and still arm because its adversary arms. A company may prefer a slower research race and still accelerate because investors compare it to competitors. A teenager may know that social media is corrosive and still participate because exclusion has social cost. A scientist may dislike publication metrics and still optimize for them because employment is filtered through them.

Status, Comparison, and Escalation

The system produces hunger even where persons resist it.

This gives us a first rigorous definition of civilizational appetite. It is not the sum of private cravings. It is the set of feedback loops that make expansion, acquisition, or optimization necessary for actors who wish merely to preserve their position.

A person can step off a treadmill.

A civilization can make the treadmill the floor.

Efficiency offers another example. The intuitive story says that if a technology uses fewer resources per unit of service, total resource use should fall. Often it does. But rebound effects occur when lower effective cost induces more use or when savings are spent elsewhere. Reviews of energy behavior show that rebound can erode expected savings; the magnitude varies by context and is a matter of empirical analysis, not a universal law. [18][30] A 2026 study of rooftop solar in Europe, for example, found evidence that households may increase electricity consumption after adoption when the marginal electricity feels cheaper or "free." [32]

The lesson is not that efficiency is pointless. It is that **capacity and restraint are different technologies.**

An efficient engine does not decide how far to drive. A cheap solar electron does not decide whether to conserve. A powerful model does not decide whether a problem should be automated. A gene-editing tool does not decide which edits belong outside the domain of acceptable action.

For most of modern history, civilization behaved as though better engines would eventually solve the steering problem.

Comparison turns private desire into a multiplayer game. In a single-player world, a person can decide that a house, income, level of security, or amount of attention is sufficient. In a competitive social environment, another person's gain can alter the meaning of one's own position even when nothing material has been lost. This is not merely vanity. Relative position can influence access to mates, networks, education, political voice, safety, and future opportunity. Status is symbolic, but symbols can carry real consequences.

The civilizational danger is that relative competition has no natural aggregate finish line. If everyone doubles the resources spent signaling prestige, relative rank may remain almost unchanged while the material and psychological cost of the competition doubles. The structure resembles an arms race: each participant can have a reason to continue even when all participants would prefer a world in which the race were cheaper.

Red Queen dynamics capture this logic. In evolutionary and organizational settings, actors may need continuous improvement simply to preserve relative position. Running faster does not necessarily move the runner forward; it prevents falling behind. Modern states encounter this in military capability. Firms encounter it in research, marketing, data accumulation, and automation. Individuals encounter it in education, credentials, visibility, and consumption. The resulting appetite is

not produced only by greed. It is produced by fear of relative decline.

This is why exhortations to “want less” often fail when aimed only at individuals. The person who unilaterally exits a status race may pay a real price. Durable restraint therefore requires institutions that change the game: norms that make some competitions less prestigious, rules that limit destructive escalation, public goods that reduce the penalty for falling behind, and forms of coordination that make mutual restraint credible. Wisdom is not merely an internal virtue when the environment rewards the opposite virtue.

Status competition also explains why moral condemnation is often ineffective. People do not merely imitate the rich because they are foolish; they read status signals as information about what a society rewards. When prestige attaches to conspicuous consumption, relentless work, technological dominance, or public visibility, imitation becomes a form of cultural learning. Changing appetite therefore requires changing prestige. A society that praises moderation in speeches while allocating admiration to excess teaches the opposite lesson with greater credibility. The most powerful curriculum is the distribution of honor.

More information would make us wiser. More productivity would make us satisfied. More connectivity would make us less lonely. More weapons would make us safer. More choice would make us freer. More medicine would make us healthier. More communication would make us understand one another.

Sometimes these promises were substantially true. Sometimes the benefit itself created the next dependence. The connected person became reachable at all times. The informed citizen became saturated. The safer state acquired more catastrophic means of remaining safe. The healthier population acquired more years in which chronic disease could emerge and more medical possibilities around which expectations could expand.

The hunger is not evidence that progress was fake. It is evidence that progress does not contain its own stopping rule.

This distinction is where both naive techno-optimism and naive anti-modernism fail. The first assumes that the next increase in capability will resolve the pathologies created by capability. The second assumes that capability itself is the pathology. The harder conclusion is that capability needs a governor. Biology discovered this before civilization existed. We call the failure cancer.

Chapter 6 — Cancer: The Biology of Broken Limits

How Multicellular Life Disciplines Cells

Cancer is frightening partly because the enemy is made of us.

A malignant cell is not a foreign invader in the usual sense. It descends from the organism's own cells. It uses ordinary biochemical machinery. It consumes glucose, oxygen, amino acids, signaling pathways, blood vessels, and space. Many capabilities that make tumors dangerous are distorted versions of capacities life requires: proliferation, survival, migration, adaptation, cooperation, repair.

Cancer is not alien life.

It is life that has escaped the terms of multicellular membership.

Modern cancer biology describes a family of capabilities associated with malignancy, including sustained proliferative signaling, evasion of growth suppressors, resistance to cell death, replicative immortality, angiogenesis, invasion, and metastasis, alongside enabling characteristics and interactions with the tumor microenvironment. [12] Evolutionary work reframes these traits as a breakdown of cooperation required by multicellular life. One influential framework describes cooperation around proliferation control, regulated cell death, division of labor, resource allocation, and maintenance of the extracellular environment; more recent work extends the analysis toward the maintenance of appropriate spatial relations. [13][14]

The language is almost uncomfortably political. A healthy cell does not maximize itself. It accepts limits on reproduction. It specializes rather than attempting every function.

The cancer analogy becomes useful only after we understand the biological achievement it depends upon. Complex multicellular life is not just a large pile of cells. It is a negotiated order in which cells accept restrictions that a free-living ancestor would not. Research in evolutionary cancer biology describes several foundations of multicellular cooperation: regulation of proliferation, controlled cell death, resource allocation, division of labor, maintenance of shared extracellular conditions, and, in newer formulations, maintenance of appropriate proximity and tissue organization [13][14].

Cancer biology is useful here because it makes the architecture of cooperation concrete. Complex bodies are possible only because cellular opportunities are constrained in multiple, overlapping ways.

Table 6.1 — Cooperation Before Growth

The cancer analogy is about constraints on local advantage, not about treating people as cells.

Multicellular discipline	Cancer-associated breakdown	Civilizational question
Proliferation inhibition	Sustained proliferation	Which organizations are allowed to expand without a stopping condition?
Controlled cell death	Resistance to cell death	Which obsolete institutions or products are unable to terminate?
Resource allocation	Capture and metabolic reprogramming	Can one subsystem appropriate resources needed by the whole?
Division of labor	Loss of differentiated function	Does growth displace the function an institution was created to perform?

Multicellular discipline	Cancer-associated breakdown	Civilizational question
Shared environment	Tissue remodeling	Does local success degrade common infrastructure or ecology?
Proximity and position	Invasion and metastasis	What happens when activity escapes the boundary within which it was governable?

The right-hand column is intentionally phrased as questions rather than equivalences. There is no literal social counterpart to a cancer hallmark. The value lies in asking what higher-order function is lost when lower-level expansion becomes the dominant criterion.

Each foundation is a brake on local advantage. A cell cannot divide whenever nutrients permit. It cannot indefinitely refuse death. It cannot seize all available resources. It differentiates into a role that may prevent it from reproducing independently. It remains where the architecture of the tissue requires. It participates in a body whose success is defined at a level above the cell.

This should feel astonishing. A human body contains lineages of cells that, in a crude evolutionary sense, could gain by defecting from these constraints. Yet organisms evolved policing, signaling, immune surveillance, developmental programs, and forms of programmed cell death that make large cooperative bodies possible. The organism is powerful precisely because its components are not allowed to maximize every local opportunity.

Cancer is not one failure but a family of evolving failures in this cooperative order. Hanahan's updated hallmarks include sustained proliferative signaling, evasion of growth suppressors, resistance to cell death, replicative immortality, access to vasculature, invasion and metastasis, metabolic reprogramming, and immune evasion. These are biological mechanisms, not moral categories. But at the level of system architecture they reveal a principle of extraordinary generality: complexity depends on constraints that prevent lower-level units from converting all available opportunity into private expansion.

It receives resources through a coordinated system rather than capturing everything nearby. It stays where it belongs. When damaged beyond repair, it may die. The body's astonishing complexity is built from units that surrender options.

This is not mystical altruism. Cells do not deliberate about the common good. The constraints are products of evolution at multiple levels. Organisms in which cellular defection was insufficiently controlled were less viable. The emergence of complex multicellularity required mechanisms that suppress destructive cell-level advantage. [13][15] Large and long-lived animals are especially interesting because, in principle, more cells and more time should create more opportunities for cancer. Peto's paradox points to the evolution of diverse suppression strategies that allow large bodies to exist without cancer risk scaling naively with cell number. [15]

A whale is possible because growth evolved brakes. That sentence should be written above the entrance to every institution that worships scale.

But the cancer analogy must be handled with discipline. Societies are not organisms. Individuals possess moral status that cells do not. Political regimes have repeatedly used biological metaphors to justify persecution, ethnic cleansing, forced

sterilization, and the elimination of dissent. To call a group a cancer is to prepare language for atrocity.

So this book will not call groups cancer.

It will call **growth logics** cancerous when they display a specific structural resemblance: local expansion that undermines the system-level conditions on which expansion depends.

The object of the analogy is behavior, not people.

Consider the tumor's relationship to blood supply. A growing tumor can induce angiogenesis, recruiting the body's vascular infrastructure into service of the tumor. [12] The tumor does not merely consume existing resources; it reshapes allocation so that more resources flow toward continued growth.

Civilizations do this institutionally.

An industry can become large enough that public policy must preserve it because too many jobs, pensions, suppliers, loans, and regions depend on it. A military program creates specialized constituencies and facilities that justify continuation. A digital platform becomes infrastructure, after which regulating it threatens users who depend on it. A financial asset class grows until its collapse becomes a systemic risk, turning private scale into public obligation.

What Cancer Actually Teaches—and What It Does Not

Growth recruits circulation.

Now consider apoptosis, regulated cell death. Multicellular life benefits from the capacity to eliminate cells that are damaged, dangerous, or no longer required. Cancer cells often acquire resistance to cell death. [12] Organizations have an analogous problem without an analogous solution. Programs persist after purpose. Agencies accumulate mandates. Companies kept alive by subsidies or cheap capital continue consuming resources.

Laws remain after context. Technologies become too embedded to retire cleanly.

Civilization is better at birth than death.

We celebrate founding, launching, scaling, expanding, creating. We have fewer rituals for ending a successful thing because its time is over.

The absence of institutional apoptosis produces what might be called **civilizational necrosis by accumulation**: not death from one malignant actor, but loss of flexibility as obsolete structures remain metabolically active.

The analogy sharpens further around differentiation. Cancer is often associated with loss or disruption of specialized cellular identity. [12][13] In healthy multicellularity, a liver cell does not decide to become an eye. Complexity depends on division of labor and reliable interfaces. Modern institutions, by contrast, are often rewarded for mission expansion. Universities become real-estate developers, health systems become financial conglomerates, technology firms become media systems, media systems become political infrastructure, governments become data platforms, platforms become quasi-governments.

Sometimes integration is efficient. Sometimes every organ wants to become the body.

And then there is metastasis: movement beyond the place where a cell's behavior can be contained by local tissue architecture. A successful malignant clone colonizes new environments. Again, the political metaphor must not be applied to peoples. But it can describe an economic logic that enters every domain and translates each into the same optimization language. Attention becomes inventory. Friendship becomes engagement. Sleep becomes performance input. Childhood becomes a market segment. Genetic information becomes an asset. Grief becomes content. The sacred grove becomes timber.

The danger of cancer is not growth in itself.

The analogy between cancer and civilization is powerful enough to be dangerous. Political regimes have repeatedly described minorities, dissidents, or social groups as parasites, infections, or tumors. That language can prepare the imagination for dehumanization and violence. Nothing in this book licenses the idea that persons are cells or that a state should behave like an immune system against inconvenient citizens.

The defensible analogy runs in the opposite direction. It concerns system design, not biological identity. Cancer biology demonstrates that when components of a cooperative system acquire persistent advantages by evading constraints on proliferation, death, resource use, position, and collective function, those local advantages can damage the higher-order system on which the components themselves depend. The lesson is not “remove defective people.” The lesson is “do not design institutions that systematically reward defection from the conditions of collective survival.”

This distinction allows us to ask harder questions without pseudo-biology. What is the social equivalent of proliferation inhibition? It is not birth control imposed by a state. It may be a rule that prevents an institution from expanding indefinitely merely because expansion is administratively possible. What is the equivalent of controlled cell death? It is not killing. It may be the willingness to terminate obsolete agencies, products, subsidies, or business models rather than preserving them forever. What is resource allocation? It is the recognition that a system cannot remain healthy when one subsystem captures resources needed for the maintenance of the whole.

The analogy also reminds us that anti-cancer mechanisms are costly. Bodies spend energy on repair, surveillance, redundancy, and apoptosis. Perfect efficiency would be lethal. The same may be true institutionally. Safety margins, independent oversight,

duplicated capacity, moratoria, and constitutional prohibitions can look wasteful when nothing is going wrong. Their value becomes visible precisely when local optimization would otherwise outrun the system's ability to absorb it.

Children grow. Wounds heal. Immune cells proliferate. Organs regenerate. The danger is growth severed from organism-level feedback.

This produces a much stronger civilizational question than "Should economies grow?" The word *growth* hides too much. Growth of what? At whose cost? Measured how? For how long? Does the activity strengthen the conditions that sustain it, or does it liquidate them? Does it increase the ability of the whole system to adapt, or merely enlarge the part that is currently winning?

A tumor can have spectacular quarterly growth. That fact contains no argument for the health of the patient.

The opposite of growth is not death. In a living system, the opposite of uncontrolled growth is form.

Form is what limits make possible. A heart is useful because it is not everywhere. A tree grows because some cells stop dividing and become wood. A brain functions because neurons obey constraints on firing, wiring, and chemistry. A body survives because cells surrender the right to pursue every available advantage.

The modern civilizational fantasy is that intelligence can substitute for surrender.

We will calculate the limit rather than respect it. We will optimize near the boundary rather than leave margin. We will engineer around scarcity rather than reduce appetite. We will monitor every failure mode and therefore need no taboo. Cancer biology offers an unfashionable reply. Complexity is not what

happens when constraints disappear. Complexity is what constraints make stable.

Chapter 7 — The Civilizational Tumor

Local Rationality, Collective Ruin

If civilization were literally a body, politics would become monstrous.

That is why it is not enough to say "society is an organism" and proceed. The metaphor becomes useful only when stripped of authoritarian temptation and reduced to a systems question: **what prevents lower-level optimization from destroying higher-level viability?**

In multicellular life, the answer is encoded in evolved control mechanisms. In civilization, the answer is unstable and contested.

Markets impose some limits. Law imposes others. Religion, custom, reputation, professional ethics, family obligations, unions, constitutions, property rights, democratic elections, technical standards, fiduciary duties, scientific norms, international treaties, and plain shame all constrain what actors can profitably or legitimately do. These systems often conflict. None is sufficient. Together they form something like a distributed tumor-suppression architecture.

The modern mistake is to think of constraints only as friction.

Friction slows engines, so engineers reduce it. Regulation slows deployment, so entrepreneurs route around it. Deliberation slows decisions, so executives streamline it. Privacy slows data access, so platforms encourage consent. Due process slows punishment, so frightened publics demand exceptions. Scientific skepticism slows exciting claims, so institutions create fast tracks. Each complaint can be valid. Every safeguard can become bureaucratic theater.

But a system optimized to eliminate all friction is often a system optimized to eliminate its own immune system.

A civilization can become self-destructive without containing a single actor who prefers destruction. This is one of the most important ideas in the book because it prevents moral simplification. The farmer who pumps groundwater to save a crop, the household that buys a larger vehicle for safety, the company that automates to remain competitive, the government that subsidizes strategic industry, and the military that expands capability in response to a rival can each have reasons that are intelligible within a local frame. The higher-order harm appears only after many such choices interact.

This is the structure of a social dilemma. Each participant receives a meaningful share of the benefit from acting while sharing only a fraction of the collective cost. Restraint, if unilateral, can feel like self-sacrifice for a public good that others may consume without reciprocating. The obvious response is to coordinate, but coordination is difficult when actors disagree about facts, distrust one another, face different time horizons, or can secretly defect.

The cancer metaphor helps because it keeps our attention on scale. A proliferating cell may be locally successful and organismically disastrous. Likewise, an economic sector can be productive by its own metrics while degrading conditions on which the broader economy depends. A political strategy can win an election while corroding the information environment required for legitimate elections. A platform can maximize engagement while exhausting attention. Local success and system health are different variables.

The conceptual mistake is to infer the second from the first. Civilizations often treat accumulated local victories as proof of collective progress because local victories are easier to measure. Revenue rises, throughput increases, output grows, capability

expands, and the dashboard turns green. The substrate—soil fertility, public trust, cognitive attention, ecological resilience, institutional legitimacy—may deteriorate more slowly and in units that do not fit the dashboard. By the time the aggregate loss becomes visible, actors have adapted their lives and institutions to the very behavior causing it.

Infrastructure can lock this pattern in place. A city designed around long car commutes makes individual low-carbon choices costly. A pension system dependent on continuously rising asset values creates political sensitivity to growth. A research ecosystem built around short funding cycles rewards visible novelty. Once capital, careers, and daily routines are organized around an appetite, restraint is no longer simply a new preference; it becomes a transition problem. The later civilization waits, the more of its own structure must be rebuilt in order to stop doing what it has learned to call normal.

This is where the analogy with cancer becomes diagnostic rather than rhetorical.

A healthy society needs proliferation — new firms, ideas, communities, technologies, art, experiments. It also needs growth suppression — antitrust, solvency rules, ecological ceilings, conflict-of-interest rules, term limits, standards, and social norms that prevent successful entities from converting success into unlimited permission.

It needs survival — continuity, archives, infrastructure, traditions. It also needs apoptosis — bankruptcy, repeal, sunset clauses, institutional closure, the ability to end programs and platforms that have outlived their purpose.

It needs resource allocation — capital, food, energy, attention, care. It also needs mechanisms preventing monopolization of circulation by actors large enough to capture the allocator.

It needs mobility — trade, migration, ideas crossing borders. It also needs boundaries that keep every successful logic from colonizing every domain.

It needs innovation. It also needs the possibility that a thing can be technologically impressive and still forbidden. The tension is permanent.

That is what makes the civilizational tumor problem harder than cancer biology. A body has one evolutionary interest at the organism level: survive and reproduce. A society contains persons with different values and legitimate conflicts. There is no uncontested definition of systemic health. One person's tumor-suppression mechanism can be another's instrument of domination.

This ambiguity is not a reason to abandon limits. It is a reason to make limits legible and contestable.

The worst systems combine appetite with obscurity. They allow expansion while hiding who benefits, who pays, what thresholds exist, and how decisions can be reversed. They produce what might be called **metastatic governance**: decisions move outward faster than accountability can follow.

Competition as a Machine for Removing Brakes

Consider global supply chains. They are miracles of coordination. They also allow a consumer to receive the benefit of labor conditions, mining practices, water use, and emissions that remain outside immediate sight. Consider digital platforms. They enable speech and coordination at unprecedented scale. They also make it possible for a small number of optimization systems to shape attention across entire societies. Consider sovereign debt. It allows states to invest through time and stabilize shocks. It also allows political benefits to be brought

forward while some costs are assigned to people who cannot yet vote.

The common feature is not evil.

It is distance.

Cancer creates distance too. A cell's local success is measured against its immediate microenvironment; the organism-level death lies outside the cell's "concern" because cells have no concern. The analogy becomes morally relevant when human systems create institutional distances so large that actors can behave *as if* they had no concern.

The executive sees a cost center. The model sees an objective function. The minister sees an aggregate. The investor sees risk-adjusted return. The consumer sees a price.

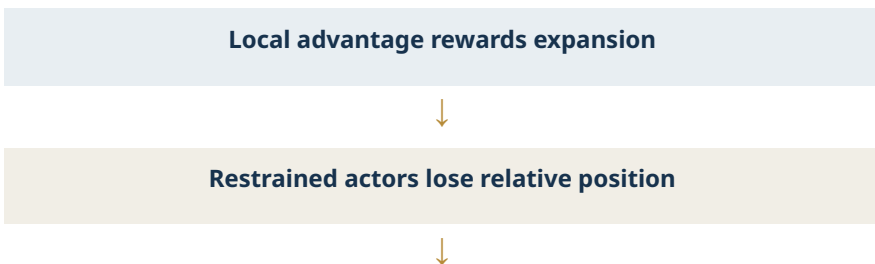
The person at the other end may see a poisoned river, an unaffordable drug, a closed factory, a flooded village, or simply no dinner.

There is no need to accuse every participant of cruelty. The system can manufacture moral anesthesia.

The sequence below is the systems core of the civilizational-tumor argument. It describes a selection environment, not an accusation about individual motives.

Figure 7.1 — Selection Against Restraint

How competition can remove brakes without any participant choosing collective ruin.



Aggressive strategies become more common



Norms adapt to the surviving strategies



System-level resilience weakens

Intervention is therefore possible at several points. Rules can change the payoff to externalization, coordination can reduce the penalty for mutual restraint, and prestige can be attached to maintenance rather than only to expansion. The loop is powerful, but it is not metaphysically inevitable.

This is why the Wendigo metaphor and the cancer analogy belong together. Wendigo gives the moral image: appetite becoming cannibalism. Cancer gives the systems mechanism: local advantage escaping higher-order restraint.

The combination yields a disturbing possibility. A civilization can become predatory without being governed by predators. It can become self-destructive without a death wish.

Competition can discipline power, improve products, expose incompetence, and prevent monopolies from becoming complacent. This book is not an argument against competition. It is an argument against treating competition as if it automatically selects for the traits a civilization needs to survive. Selection rewards whatever wins under the current rules, not whatever preserves the arena in which the game is played.

If a market rewards firms that externalize environmental costs, firms with stronger ecological restraint can be selected against. If political competition rewards outrage, actors who refuse outrage can lose visibility. If military rivalry rewards rapid

deployment, states that maintain large safety margins can feel strategically vulnerable. If scientific careers reward positive findings and publication volume, patient replication and null results can become professionally expensive. None of these patterns proves that competition is bad. They prove that the objective function is never neutral.

The dynamic can become self-reinforcing. A restrained actor exits or loses. The surviving population contains a larger proportion of aggressive strategies. Norms then shift to describe those strategies as realistic, professional, or inevitable. Institutions are redesigned around them. New entrants learn that success requires the same behavior. What began as an option becomes a baseline. The system has not merely tolerated appetite; it has trained for it.

This is a more precise version of the civilizational-tumor metaphor. The danger is not simply expansion. Healthy tissues grow when growth is needed. Healthy economies expand productive capacity when it improves lives. The dangerous condition is selection that progressively weakens higher-order constraints. A mature civilization must therefore regulate not only actors but selection environments. It must ask which kinds of institutions survive under its rules and whether those are the institutions it would deliberately choose if long-term system health were the criterion.

This is why rule design matters more than moral sorting. The aim is not to identify “greedy” firms or “bad” nations and expel them from the moral community. It is to alter the payoff landscape so that long-horizon behavior can survive. When pollution is costly to emit, safety shortcuts create liability, dangerous research requires independent review, or resource extraction is bounded by credible collective rules, restraint becomes compatible with continued participation. The civilizational task is to make wisdom selectable.

It can produce cruelty through optimization, not sadism. And because no one intends the whole, everyone can point to the innocence of the part. "My transaction is tiny." "My flight changes nothing." "My fund follows the market." "My company obeys the law." "My country cannot disarm first." "My model only recommends."

"My department does not set the budget." Each sentence can be true. The body can still die.

This is the point at which moral language must become systemic without becoming evasive. If everyone is merely a component, responsibility dissolves. If every component is personally blamed for the whole, responsibility becomes absurd. Mature responsibility lies between: design systems in which ordinary local rationality does not reliably aggregate into catastrophe.

That is precisely what taboo once attempted in crude form. It created rules that did not need to be recalculated at every encounter. Do not eat this. Do not enter there. Do not cut that tree. Do not hunt in this season. Do not touch the dead. Do not take more than this. The modern mind hears ignorance. Sometimes it is right. Sometimes it is hearing an immune system it has not yet learned to name.

Chapter 8 — The Starved Outside the Feast

Abundance Without Access

A civilization should be judged not only by what it can produce, but by what it can watch people lack.

This is an unpleasant criterion because it ruins the innocence of abundance.

The world produces more food than past civilizations could have imagined. It can move strawberries across oceans in winter, refrigerate meat across continents, optimize warehouse inventories in real time, and deliver restaurant meals to a door through a phone. Yet the 2026 *State of Food Security and Nutrition in the World* report estimated that about 645 million people experienced hunger in 2025. [19] UNEP's latest comprehensive food-waste assessment estimated that 1.05 billion tonnes of food were wasted at retail, food-service, and household levels in 2022, about a fifth of food available to consumers at those stages. [20]

These statistics should not be turned into a dishonest equation. The existence of waste does not prove that hunger would disappear if rich households cleaned their plates. Hunger is produced by poverty, conflict, displacement, failed distribution, market access, infrastructure, climate shocks, governance, and power. Food discarded in Zurich is not a logistical substitute for food missing in a besieged city.

But the juxtaposition still matters. It tells us that civilization's hunger problem is not simply "not enough food." It is also a problem of allocation, access, resilience, and what counts as a claim on abundance.

The same is true of medicine. WHO and the World Bank reported in 2025 that 4.6 billion people still lacked full coverage

of essential health services in 2023, while 2.1 billion people experienced financial hardship due to out-of-pocket health spending in 2022. [21] At the same time, biomedical science is discussing gene therapies, individualized cancer vaccines, organ engineering, brain-computer interfaces, anti-aging interventions, and AI-designed molecules.

The most morally disturbing feature of the Wendigo metaphor appears before civilization harms itself. It appears when abundance and deprivation coexist. The latest FAO assessment available in 2026 estimated that about 645 million people faced chronic hunger in 2025 and roughly 2.1 billion experienced moderate or severe food insecurity. [19] At the same time, the latest comprehensive UNEP food-waste estimate found about 1.05 billion tonnes of food waste at household, food-service, and retail levels in 2022, close to one fifth of food available to consumers at those stages [19][20].

A table can make juxtaposition look like causation, so the methodological warning must be explicit. These indicators come from different systems, years, and definitions. They are placed together to show coexistence, not to claim a single mechanism.

Table 8.1 — Feast and Deprivation in the Same World

The coexistence is morally significant, but causal claims must remain disciplined.

Indicator	Latest global estimate used here	What the number does — and does not — show
Chronic hunger	~645 million people in 2025 (SOFI 2026)	Persistent deprivation despite immense productive capacity; not proof that food waste can simply be redirected person-for-person.
Moderate or severe food insecurity	~2.1 billion people in 2025 (SOFI 2026)	Large vulnerability in access; categories are

Indicator	Latest global estimate used here	What the number does — and does not — show
		broader than chronic undernourishment.
Food waste	1.05 billion tonnes in 2022 at household, food-service and retail stages (UNEP 2024)	Large avoidable waste; includes inedible parts and does not by itself identify the causes of hunger.
Essential health-service coverage	~4.6 billion not fully covered in 2023 (WHO/World Bank 2025)	Uneven access to basic health services; not an argument against frontier biomedical research.
Financial hardship from health spending	~2.1 billion people in 2022 (WHO/World Bank 2025)	Shows access can fail through affordability even where treatment exists.

The structural question is whether a civilization that can organize extraordinary abundance also builds institutions capable of translating abundance into reliable access. A system should not be allowed to call scarcity “natural” merely because access failures are administratively complex.

These numbers must not be paired carelessly. Food discarded in one city cannot simply be teleported to a household in a war zone. Hunger is driven by conflict, poverty, price shocks, infrastructure, displacement, governance, and access as well as by production. Food waste includes inedible fractions. The coexistence of hunger and waste is not proof that a single warehouse contains the missing meals of humanity.

But the coexistence does refute a simpler moral story: that deprivation persists merely because humanity lacks the technical

capacity to produce enough. In many domains, the problem has become allocation, purchasing power, institutional reach, political priority, and the protection of people who cannot successfully bid for what they need. A market can be extremely efficient at serving solvent demand while remaining silent about need that lacks money.

This is where Kimmerer's modern use of the Windigo becomes especially sharp. In *Braiding Sweetgrass*, she describes a world in which goods can accumulate while others remain excluded, and uses “Windigo economics” to name the moral inversion by which sanctioned appetite coexists with unmet need. The deeper civilizational point is not confined to capitalism. Any allocation system can become monstrous when it treats the inability to command resources as evidence that a person's need is negligible.

The latest hunger data also contain an important reason not to indulge fatalism. Global hunger fell again in 2025. The FAO explicitly emphasizes that this progress shows hunger is not inevitable. That matters for the argument of this book: “Wendigo” should not become a mystical excuse for despair. Systems produce deprivation, and systems can reduce it. The warning is directed at complacency—especially the complacency that interprets technical abundance as evidence that access will solve itself. Decline in hunger is an achievement to protect, not evidence that the moral problem has disappeared.

Again: frontier medicine does not cause missing primary care. The moral contradiction is more subtle. A civilization may be structurally better at creating new frontiers than at distributing old victories.

There are incentives for novelty. There are patents, prestige, venture returns, prizes, headlines, publications, military advantage, and national competition. There is less glamour in making a cheap intervention reliably reach the last village, the

last migrant camp, the last poor neighborhood, the last person without insurance, the last neglected disease.

The Wendigo therefore does not merely consume too much. It can also consume **attention** in a way that leaves solvable suffering invisible. The future is exciting. Maintenance is not. The breakthrough receives a keynote. The sewage system receives a budget line. The new model is announced. The old clinic loses staff.

The moonshot is photographed. The patient waits.

There is no necessary trade-off between ambition and provision. Rich civilizations should be able to do both. The scandal is precisely that the capacity exists and the integration fails.

This is where Thanatos becomes a social role rather than a psychic drive.

The "followers of the death god" in a modern civilization are not people who love death. That accusation would be melodramatic and usually false. They are institutions that become able to treat preventable death as background noise while foregrounding a different objective.

The Moral Geometry of the Externality

A finance ministry cannot save everyone.

A hospital cannot provide every treatment.

A military planner cannot eliminate casualties.

A public-health agency cannot reduce all risk to zero.

Scarcity is real. Trade-offs are real. The danger begins when these truths become a blanket moral anesthetic, allowing avoidable suffering to be redescribed as natural simply because it is administratively inconvenient.

Death is easiest to accept when it arrives as a statistic. The starving person becomes "food insecurity." The untreated patient becomes "coverage gap." The worker becomes "labor flexibility." The displaced community becomes "transition cost." The species becomes "biodiversity loss." The future person becomes "discount rate." Abstraction is necessary for governance. Nobody can administer billions of lives through personal acquaintance.

But abstraction has a spiritual hazard: it can make the victim disappear before the harm does.

The same pattern appears in health. The WHO and World Bank reported in the 2025 Global Monitoring Report that approximately 4.6 billion people were not fully covered by essential health services in 2023, while 2.1 billion people experienced financial hardship from out-of-pocket health spending in 2022. [21] These figures coexist with spectacular progress at the frontier: precision oncology, gene therapies, AI-assisted discovery, robotic procedures, and serious research into extending healthy lifespan.

Again, the frontier does not cause the absence of basic care. Advanced research can create treatments that later diffuse widely, and rich health systems can fund discoveries that benefit the world. The ethical problem is not excellence. It is the possibility that a civilization becomes mesmerized by the next frontier while regarding the failure to distribute yesterday's frontier as administratively ordinary.

Externalization has a geography. Harms tend to travel toward people with less bargaining power, toward places with weaker regulation, toward ecological systems that cannot sue, and toward future generations that cannot vote. Benefits often travel in the opposite direction, toward actors capable of paying, owning, insuring, or influencing. This does not happen in every case, but it happens often enough that the distribution of risk

should be treated as part of the design rather than as an afterthought.

The Wendigo first eats the neighbor because the neighbor is easier to externalize than the self. The mine contaminates someone else's watershed. The carbon emission enters a shared atmosphere. The dangerous product is sold into a jurisdiction with weaker enforcement. The debt matures after the political term ends. The waste is shipped away. A civilization begins to eat itself only after it has practiced for a long time on people and places it has learned not to count fully.

The same geometry appears within wealthy societies. Average abundance can conceal households rationing heat, medicine, dental care, time, or safe housing. Aggregate indicators remain necessary, but the distribution of exposure determines whether a society experiences its own prosperity as shared infrastructure or as a feast visible through glass. A civilization becomes unstable when the excluded are continually shown the abundance they cannot reliably access, while the included are encouraged to interpret exclusion as somebody else's personal failure.

Ancient sacrificial systems at least admitted that something was being placed on an altar. Modern systems often prefer to call the altar an externality.

This chapter is not an argument for compulsory equalization of every outcome. Unequal rewards can motivate effort, reflect different preferences, compensate risk, and support innovation. Nor is every unmet need an injustice. Human desires are unbounded; no society can satisfy them all.

The hard line is elsewhere.

A civilization that celebrates its ability to manufacture desire while failing to secure basic conditions of agency for large populations should not call the resulting deprivation mysterious.

If people are hungry while edible food is discarded at vast scale, the system has an allocation pathology even if the causal channels are complex.

If essential care remains inaccessible while medical frontiers race forward, the system has a distribution problem even if frontier research is itself good.

If the benefits of growth are privatized while ecological and health costs are dispersed, the system has learned a cannibal trick: **it has made the body of the victim difficult to see.**

Windigo stories were frightening because the cannibal was not abstract. Modern civilization needs an equivalent moral compression. Not because statistics are false. Because statistics are too easy to survive emotionally.

The civilized form of cannibalism is not eating human flesh. It is building a system in which other people's preventable suffering becomes an acceptable input cost.

That sentence should offend. It is meant to.

The offense should not end in guilt. Guilt is one more private emotion that systems can absorb without changing. The useful response is to recover the category that modern optimization distrusts most: the forbidden trade. Some things should not be bought with other people's basic survival. Some efficiencies should be unavailable. Some prices should be too high even when the payer is far away. That is the territory of taboo.

PART III

THE FORBIDDEN LINE

*A civilization without taboos is not necessarily free. It
may simply have forgotten which cliffs are real.*



Chapter 9 — Why Human Beings Built Forbidden Zones

Taboo as Social Technology

The word *taboo* is itself a warning imported into modern English from Polynesian languages, where forms such as *tapu* or *tabu* refer to what is set apart, forbidden, sacred, or dangerous. Anthropological scholarship has long treated taboo as more than one phenomenon: it can protect sanctity, encode hierarchy, organize pollution beliefs, regulate diet, structure ritual, police sexuality, preserve authority, or manage relationships with resources. [23][27] There is no single theory that makes every taboo sensible.

That is exactly why modernity was right to distrust them.

Taboos can be cruel. They can stigmatize menstruation, illness, sexuality, caste, disability, outsiders, foods, occupations, and forms of thought. They can freeze accidental custom into moral necessity. They can protect kings by declaring their bodies sacred and silence criticism by making questions impure. To romanticize taboo is to romanticize a technology that has often served domination.

And yet a different mistake has become fashionable: assuming that because many taboos are irrational or unjust, the *form* of taboo has no function worth preserving.

The form is simple. A taboo is a rule stronger than ordinary preference. It places an action outside routine bargaining. You do not recalculate it every morning.

That feature can be oppressive. It can also be exactly what a society needs when incentives are strong, consequences are delayed, causal knowledge is incomplete, and one violation can be difficult to reverse.

The book cannot defend taboo responsibly without showing both sides of the architecture. Strong prohibitions can preserve fragile goods precisely because they resist negotiation, but the same resistance can shelter error and power.

Table 9.1 — Strong Boundaries and Their Failure Modes
A defense of limits must include a theory of how limits themselves become dangerous.

Boundary form	Potential protective function	Characteristic failure mode
Sacred taboo	Removes a valued object or action from ordinary bargaining	Can fossilize superstition, hierarchy, exclusion, or false causal beliefs.
Constitutional right	Protects persons and domains from majoritarian optimization	Can become rigid or be interpreted selectively by powerful actors.
Scientific / safety moratorium	Buys time where uncertainty and irreversibility are high	Can be captured by incumbents or persist after evidence changes.
Ecological closure / rāhui-like restriction	Temporarily limits harvest or access so a resource or place can recover	May be romanticized or misapplied outside its cultural and ecological context.
Technical interlock / operating envelope	Makes unsafe states harder or impossible to enter accidentally	Can be bypassed when performance pressure rewards circumvention.

A modern wisdom of limits therefore needs legitimacy, evidence, contestability, and periodic review. The goal is not to make boundaries weak. It is to make strong boundaries corrigible without making them trivially negotiable whenever appetite becomes impatient.

Modern engineering has rediscovered this logic under less embarrassing names. Interlocks. Safety margins. Air gaps. Hard limits.

The word *taboo* entered European languages through encounters with Polynesian concepts such as *tapu*, but the broader phenomenon is nearly universal: communities mark certain acts, places, foods, persons, times, or relationships as prohibited in a way that exceeds ordinary preference. Modern secular societies often hear the word and think first of irrationality. That suspicion is justified often enough. Taboos can encode prejudice, hierarchy, misogyny, caste, fear of disease, or causal beliefs that are simply false.

Yet a taboo also performs an institutional function that modern policy sometimes struggles to reproduce. It reduces decision frequency. Instead of recalculating from first principles every time a temptation appears, a community removes an option from the ordinary choice set. This can be valuable when the short-term benefit of violation is salient and the long-term cost is uncertain or collectively distributed.

Anthropological research does not support the romantic claim that taboos are secretly perfect conservation laws. A 2023 systematic review found many cases in which food taboos influenced wildlife use, but effects vary by species, context, enforcement, and cultural meaning. [24] Research on Pacific fisheries similarly cautions against treating customary restrictions as if they were designed from modern ecological science. Still, practices such as Māori *rāhui* show that culturally authoritative prohibitions can function as temporary closures that allow resources or places to recover. [26]

The important lesson is architectural rather than nostalgic. Human groups have long used emotionally and socially powerful “do not” rules to protect values that would be vulnerable if every encounter reopened negotiation. Modern civilization did not

outgrow this need. It changed the vocabulary. Constitutional rights, research ethics prohibitions, protected areas, conflict-of-interest rules, moratoria, and safety standards are secular descendants of the same design problem: some options must be made difficult to exercise precisely because they will sometimes be tempting.

Separation of duties. Two-person rules. Fail-safe defaults. No-fly zones. Sterile fields. Quarantine. Nuclear exclusion zones. Protected habitats. Constitutional rights. Some constraints are intentionally made difficult to optimize away.

We do not ask a pilot to recalculate from first principles whether the redline on an engine gauge is metaphysically justified. We do not let every surgeon decide whether sterility is worth the inconvenience. We do not ask whether a bridge's safety factor maximizes today's traffic throughput. We encode historical knowledge, including knowledge gained from failures, into rules that ordinary operation is not allowed to bargain with.

Civilization already believes in secular taboos. It merely dislikes the word.

The ecological literature offers cases where customary prohibitions can function as resource-management tools, though the evidence also warns against simplistic stories that every traditional taboo evolved "for conservation." [24][25] Māori practices such as *rāhui* can temporarily prohibit access to resources or places and have been used in contemporary resource management, including closures that allow depleted fisheries or shellfish beds to recover. [26] The important point is not that customary practice should replace ecological science. It is that societies long ago discovered a social technology modern regulators still need: **temporarily remove a resource from ordinary appetite.**

There is wisdom in the grammar. Not "the price is higher." Not "please consume responsibly." Not "consider future generations." No. You do not take from here now.

Markets are extraordinary coordination systems, but price is a weak guardian for things whose loss is irreversible, whose value is contested, or whose future scarcity is not represented by present ownership. If an old-growth forest can be destroyed and the resulting profit is larger than the legally recognized cost, the price has not proven the forest should fall. It has proven that the accounting boundary was drawn in a way that permitted the conversion.

When the Reason Is Forgotten

Taboo changes the burden of proof.

In an ordinary market, use is permitted unless prohibited.

In a taboo domain, use is prohibited unless released.

That inversion matters for technologies as much as ecosystems.

Imagine a class of biological experiments with a plausible path to catastrophic misuse. The default modern instinct is often to regulate conditions of access, create review procedures, and proceed if compliance can be demonstrated. A taboo architecture would ask a prior question: should this capability exist outside a tiny, explicitly authorized domain at all?

Imagine an AI system with the ability to autonomously replicate, acquire resources, evade shutdown, and conduct high-consequence operations. A permissive architecture asks how to deploy safely. A taboo architecture asks why these capabilities should be composed in the first place.

Imagine surveillance so comprehensive that it could prevent many crimes. The optimization logic asks how to balance privacy and safety. A taboo logic may begin by declaring certain forms of

monitoring categorically incompatible with free life, even if measurable safety is sacrificed.

The reader should feel the danger on both sides. Categorical prohibitions can preserve liberty. Categorical prohibitions can also destroy it. So what distinguishes a wise taboo from a tyrannical one? Not age. Not religion. Not majority approval. Not tradition by itself.

A defensible modern taboo should protect conditions of agency, reciprocity, or system viability against incentives that would otherwise predictably erode them. It should be explicit about what it protects. It should be subject to rare but real revision. It should not convert disfavored persons into forbidden objects. And it should be hardest to relax precisely when the short-term rewards for relaxation are highest.

That last condition is crucial. A safety mechanism that disappears under pressure is not a safety mechanism.

Taboos have a serious failure mode: the rule can outlive the reason. Ze Hong's 2024 account [23] of the cognitive origin and cultural evolution of taboos emphasizes that prohibitions may arise through imperfect causal inference and then be transmitted after their original rationale has become vague or lost. This helps explain why taboos can accumulate, mutate, and survive as conventions whose consequences are poorly understood.

That history gives modern skepticism its force. A rule that cannot explain itself becomes vulnerable to manipulation by authority. "Because it is forbidden" can protect a wetland, but it can also protect a tyrant. A civilization that respects every inherited prohibition would freeze injustice into sacred form. The task is therefore not to restore taboo as obedience. It is to recover the functional category of a strong boundary while subjecting the content of that boundary to evidence, rights, revision, and public reasoning.

This creates a difficult asymmetry. If we demand complete causal proof before protecting anything, fragile systems may be destroyed before certainty arrives. If we allow unexplained prohibitions to dominate indefinitely, error becomes untouchable. A mature approach needs both precaution and sunset: the willingness to establish a protective constraint under uncertainty, and the willingness to revisit it when knowledge changes.

The lost rationale of old taboos also contains an epistemic warning for modern technocracy. We often inherit institutions whose purpose is no longer obvious because they succeeded. Redundancy looks wasteful after decades without failure. Separation of powers looks slow when institutions are stable. Strict research rules look bureaucratic when accidents are rare. Ecological buffers look like unused land. The absence of disaster can erase memory of why the brake existed. A civilization can therefore dismantle its restraints not because the danger vanished but because the restraint made the danger invisible.

The taboo against torture is tested when torture appears useful. The taboo against political assassination is tested when the opponent seems dangerous. The taboo against human experimentation without consent is tested when knowledge seems urgent. The taboo against first use of certain weapons is tested when war turns desperate.

The taboo against consuming breeding stock is tested during famine. If a prohibition binds only when it costs nothing, it is etiquette.

Ancient cultures surrounded limits with gods, impurity, ancestors, and fear partly because naked prudence is weak against immediate appetite. The sanction had to be emotionally larger than the temptation.

Modernity removed the gods. It has not removed temptation. The question is whether law, culture, professional identity, and

institutional design can generate constraints with comparable force without restoring superstition. This book calls them **sacred brakes**. The adjective is intentionally uncomfortable.

A sacred brake is not sacred because a priest says so. It is sacred because the system chooses to place some condition of continued life beyond routine optimization.

The atmosphere can be such a condition. Human bodily autonomy can be. The right not to be tortured can be. A nuclear command chain can contain sacred brakes. A scientific field can maintain sacred brakes around experiments that create disproportionate irreversible risk. An economy can treat the subsistence floor beneath human life as a sacred brake against pure market allocation. The objective is not to make more things holy. It is to remember what holiness once did technically: **it created a boundary appetite was not authorized to price.**

Chapter 10 — When Limits Became Insults

Modernity and the Liberation from False Boundaries

Modernity learned to hear the word *limit* as a challenge.

There is grandeur in this.

The limit of the horizon became navigation. The limit of gravity became flight. The limit of infection became antibiotics and vaccines. The limit of night became electric light. The limit of distance became telecommunication. The limit of computation became a moving target that entire industries learned to push every year.

A species surrounded by death became a species that could say: not yet. Any book defending limits must bow before this rebellion. The problem is not that humanity fought limits. The problem is that success trained us to stop distinguishing kinds of limits. Some limits are barriers imposed by ignorance.

Some are barriers imposed by unjust power. Some are engineering constraints waiting for a better design. Some are trade-offs that can be shifted but not abolished.

The modern suspicion of limits did not appear from nowhere. It was earned through centuries of conflict with boundaries that claimed sacred or natural authority while protecting privilege. Hereditary rank was called order. The exclusion of women from education was called nature. Racial hierarchy was called science. Censorship was called protection of truth. Poverty was called fate. Disease was surrounded by prohibitions that often delayed understanding. In each case, asking “why?” and refusing “because it has always been so” could be an act of moral and intellectual courage.

Science intensified this emancipatory habit. A mountain was no longer a divine wall but geology. Infection was no longer a curse but biology. Lightning was no longer punishment but electricity. Limits that looked metaphysical dissolved into mechanisms that could be understood and sometimes manipulated. Modern technology then turned understanding into agency. An ocean became crossable, night became illuminable, organ failure became treatable, information became transmissible at planetary scale.

It is difficult to overstate the civilizational value of this achievement. A culture that simply accepted the first boundary it encountered would have left vast quantities of avoidable suffering intact. The capacity to say “this limit may be contingent” is one of humanity’s great moral technologies.

The Wendigo thesis does not reject that inheritance. It asks what happens when a successful heuristic becomes universal. If many prohibitions were false, are all prohibitions presumptively false? If many limits could be moved, is every limit merely waiting for a sufficiently clever tool? If liberation often meant expanding choice, must every additional choice count as liberation? The transition from justified skepticism to reflexive boundary-hostility can occur almost unnoticed because it uses the language of past victories.

The success of boundary-breaking also changes identity. Innovators are taught to see friction as a problem to be removed. Entrepreneurs are rewarded for turning previously fixed arrangements into services. Scientists are trained to doubt received categories. Activists are trained to challenge norms that naturalize injustice. These are valuable dispositions. But a civilization composed only of boundary-breakers would resemble a vehicle containing engines and no brakes. The missing profession is not a caste of pessimists; it is the cultivation of people capable of distinguishing liberating transgression from

destructive transgression without confusing caution with obedience.

Some are boundary conditions of the system itself. And some are limits we cannot identify confidently until after crossing them. A mature civilization would develop an entire literacy around these categories. Instead we often inherit a single heroic reflex: if reality says no, innovate.

This reflex has political power because "limit" has been used dishonestly. Elites have told the poor to accept their place. Patriarchies have called female subordination natural. Empires have called domination inevitable. Economists, priests, scientists, and bureaucrats have all mistaken contingent arrangements for permanent constraints. Human emancipation has repeatedly required someone to say that the alleged law of nature is merely a rule written by beneficiaries.

So suspicion of limits is earned. But earned suspicion can become generalized incapacity.

The rhetoric of disruption reveals the shift. A domain is "ripe" for disruption when inherited practices create friction. Sometimes friction is rent-seeking and bureaucracy. Sometimes friction is memory. Sometimes it is due process. Sometimes it is professional caution created by previous disasters. Sometimes it is the slow, maddening work by which a society refuses to let one objective erase all others.

Speed cannot tell the difference. Nor can competition.

The ideology of inevitability makes matters worse. A technology will be developed somewhere, therefore we must develop it here. A market will exist, therefore banning it only transfers profit. An adversary may deploy, therefore restraint is unilateral disarmament. Consumers want it, therefore refusal is paternalism. Investors expect growth, therefore stability is failure.

Each sentence contains some truth. Together they abolish agency. The civilization most proud of freedom repeatedly describes its most consequential choices as things it has no choice but to do. This is the Wendigo's favorite grammar. "I must eat because another may eat first."

The Red Queen turns appetite into obligation. [17] A state that wants only security can enter an arms race. A company that wants only survival can enter a scale race. A laboratory that wants only relevance can enter a capability race. Nobody needs to desire the destination. Each merely avoids falling behind.

The Error of Treating Every Boundary as False

At that point, the limit is experienced not as a guardrail but as betrayal.

The engineer who raises a safety concern is "blocking innovation."

The regulator is "anti-growth."

The ethicist is "not technical enough." The community defending land is "anti-development." The worker demanding slower deployment is "resistant to change." The scientist proposing a moratorium is "ceding leadership." Sometimes these accusations are fair. That is what makes them powerful. A wise limit can be indistinguishable, in the short term, from cowardice.

The myth of Icarus is often reduced to "do not fly too high," which misses its ambivalence. Flight itself is the miracle. The warning concerns losing the conditions that make flight survivable. The wings are not condemned; the failure is a relation to altitude, heat, and instruction.

Erysichthon is sharper for our purpose. He does not accidentally exceed a technical envelope. He attacks the category of the forbidden. The sacred grove says: there is a place your reasons do not authorize you to consume.

A physical boundary differs from a political prohibition, and an ecological threshold differs from a moral taboo. The danger begins when these categories are rhetorically flattened. A social reformer says that unjust traditions must be broken; an engineer says that technical obstacles can be overcome; an entrepreneur says that regulation should not block innovation. Each claim can be reasonable. Together they can create a cultural background in which “limit” becomes synonymous with “failure of imagination.”

The planetary-boundaries framework is useful here precisely because it is not a list of supernatural commandments. It identifies Earth-system processes associated with the stability and resilience of the conditions under which human civilization developed. The 2025 assessment reported seven of nine boundaries as transgressed, including ocean acidification for the first time. [22] These are not precise cliffs, and the framework is debated in details, but its logic is difficult to dismiss: some forms of pressure increase systemic risk even if technology can postpone, redistribute, or partially repair particular effects.

The mature response to a boundary is therefore neither worship nor contempt. It is diagnosis. Some limits are cruel conventions and should be abolished. Some are engineering constraints and can be moved. Some are safety margins and should be respected until better evidence exists. Some are constitutional or ethical boundaries whose value lies in protecting persons from optimization by others. Some are physical constraints that will eventually enforce themselves whether or not we acknowledge them.

A civilization that calls every restraint cowardice will learn the difference late. Reality does not negotiate merely because a culture has built its identity around transgression. The airplane door is also a door. The fact that opening doors usually enlarges

freedom does not make opening this one at altitude an act of liberation.

The most serious boundaries are often probabilistic rather than visible. A bridge load limit can be tested relatively directly. The safe operating region of a climate, ecosystem, financial network, or globally connected technological system is harder to observe because feedbacks interact and thresholds may be uncertain. Uncertainty is therefore not automatically an argument for action or inaction. It changes the burden of proof according to reversibility and scale. Where errors can be repaired cheaply, experimentation deserves freedom. Where errors may be global and irreversible, ignorance itself becomes part of the risk.

The axe says: reasons are all that matter, and mine are sufficient. This conflict now appears in secular form wherever civilization meets an irreversible frontier. How much biodiversity may be lost before ecosystem function becomes dangerously fragile? How much warming can be tolerated before feedbacks become difficult to control?

How much autonomy should be delegated to machines before meaningful human oversight becomes ceremonial? How much surveillance can a democratic society normalize before privacy survives only as a legal fiction? How much genetic intervention belongs in descendants who cannot consent? How much debt can be shifted to future citizens before present voters become political Erysichthon, consuming what is not solely theirs?

Science can inform these questions. It cannot erase the need for judgment. A boundary is not less real because its exact coordinate is uncertain.

In engineering, uncertainty usually increases the safety margin. In politics and economics, uncertainty often becomes an argument for delay in restraint: because we do not know precisely where the cliff begins, we continue walking.

That is backwards. The more irreversible the fall, the more ignorance should count in favor of margin.

This is not the precautionary principle in its laziest form, where any speculative harm can veto any action. Such a principle would freeze civilization. Risks exist on both sides. Failing to innovate kills too. Delaying medicines, energy systems, or defensive technologies can impose real harm.

The required principle is harder: **Treat uncertainty symmetrically, but treat irreversibility asymmetrically.**

When both action and inaction carry risk, compare them honestly. But when one path can close the option of reversal for everyone, the burden of proof should rise. Modernity's heroic reflex says the mountain exists to be climbed. Wisdom asks whether the climber can descend.

Chapter 11 — The Factory of Hunger

From Meeting Needs to Manufacturing Salience

Ancient appetite waited for objects.

Modern appetite can manufacture them.

This is one of the largest psychological changes in human history and one of the least metabolized morally. A subsistence economy mainly answers existing needs and culturally defined wants. An advanced consumer economy does something more sophisticated: it searches the space of possible wants, tests stimuli, measures response, segments populations, and iterates.

Desire becomes a research program.

There is nothing inherently sinister about this. Product discovery can reveal genuine unmet needs. People do not have perfect introspective access to what will help them. Entrepreneurs often create tools users could not have described in advance. Art creates desires for experiences no audience knew existed. Medicine can reveal treatable suffering previously accepted as normal.

But once desire becomes measurable, it becomes optimizable. The supermarket arranges products. The casino arranges reinforcement. The social platform arranges attention. The streaming service arranges continuation. The app arranges notification. The credit system arranges temporal access to future income. The advertising system arranges deficiency.

Economic development began, in large part, by solving obvious needs more reliably: food, shelter, clothing, transport, sanitation, medicine, communication. Markets proved astonishingly effective at coordinating specialized production and responding to demand. But once a substantial share of purchasing power moved beyond subsistence, a new commercial

problem appeared. Firms did not merely need to satisfy existing preferences; they needed to win attention among competing possibilities.

Attention is the gateway through which latent preference becomes active demand. A person may be physically comfortable and yet be made acutely aware of a difference between the life they have and the life they could display, experience, or imagine. Advertising perfected the art of making that gap emotionally vivid. Digital systems make the process measurable. They test which image, timing, comparison, or social signal moves behavior and then optimize at scale.

The phrase “manufacturing desire” can sound conspiratorial because no central planner designs the whole machine. A better description is evolutionary. Messages that fail to activate behavior disappear. Messages that reliably convert attention into action are copied, refined, and automated. Over time, the ecosystem becomes populated by increasingly effective techniques for keeping dissatisfaction salient.

This is not always harmful. A campaign can make a person aware of a vaccine, an educational opportunity, a safer product, or a service that genuinely improves life. The problem is structural: the same infrastructure that discovers needs can learn to exploit vulnerabilities, and the business model may have no endogenous reason to stop once the need has been met. The factory of hunger becomes dangerous when its success metric is continued activation rather than durable satisfaction.

The cultural consequences go beyond consumption. Political communication also competes for attention, and emotionally activating messages can outcompete nuanced ones. Scientific institutions compete for visibility, creating incentives for novelty and strong claims. Personal identity competes for recognition in public networks. The factory of hunger therefore produces not only customers but selves that must remain legible, current, and

reactive. A person may begin to experience quiet not as peace but as disappearance. That is a profound transformation in the psychology of enough.

The recommendation model arranges the next want before the previous one has cooled. The modern hungry ghost carries a supercomputer in its pocket. Its throat is still narrow.

The crucial transition is from selling objects to selling **continued engagement with appetite**. A product can satisfy and exit. A platform has stronger incentives when satisfaction becomes another state inside the loop.

Finish one video: here is another. Buy one object: people like you also bought this. Reach one fitness goal: optimize the next metric. Complete one course: earn another credential. Read one political outrage: here is a more activating one. Find a partner: now optimize the relationship. Improve the face: now notice the next defect.

Increase productivity: now schedule the recovered time. The machine does not need to make us miserable. It needs to make completion rare.

This is a subtle form of civilizational self-consumption because attention is a finite biological resource. A platform can increase total "value" as measured internally while degrading the user's ability to allocate attention elsewhere. A labor system can increase productivity while converting the saved time into more work. A communication system can reduce the cost of contact while increasing the volume of obligations. Efficiency can become a mechanism for filling every freed space.

The rebound effect is therefore larger than energy.

When an activity becomes cheaper in money, time, or effort, we often do more of it. [18][30] Digital photography made photographs nearly free and produced not a modest archive but oceans of images. Messaging reduced communication friction

and produced more messages. Computing became dramatically cheaper per operation and total computation exploded. Solar electricity can make marginal consumption feel cheaper and thereby increase use in some contexts. [32] Civilization repeatedly converts efficiency into scale. This is not a bug in markets. It is one of their triumphs. It is also why efficiency alone cannot be our theory of sustainability.

The Algorithm That Must Keep You Wanting

Now add artificial intelligence.

AI can lower the cost of producing text, images, software, persuasion, research hypotheses, entertainment, companionship, advertising, personalized education, and perhaps scientific discovery. The obvious optimistic scenario is abundance: intelligence becomes cheaper; creative production becomes accessible; services become personalized; knowledge becomes easier to obtain.

All plausible.

But the Wendigo question is different: What happens to appetite when the cost of feeding appetite collapses? If content becomes nearly free, do humans consume a fixed amount and use the saved time for silence? Or does content expand until attention becomes the limiting reagent? If personalized persuasion becomes cheap, do firms advertise less because each advertisement is more efficient?

Or does every surface become persuasive because the marginal cost is negligible? If software becomes cheap to create, do organizations stabilize their systems? Or does software proliferate until maintenance and cognitive complexity absorb the savings? If scientific hypotheses become cheap, does science become clearer? Or does verification become the new bottleneck while discovery claims flood the epistemic environment?

If synthetic companions become emotionally compelling, do people feel socially complete? Or does the standard for companionship shift toward continuous customization that human relationships cannot match?

Recommendation systems are often described as mirrors of preference, but a mirror that selects what to show next is already participating in preference formation. The system observes behavior, predicts what will retain attention, and changes the information environment in response. The user then behaves inside that changed environment, producing new data that refine the next prediction. This is not mind control. It is a feedback loop, and feedback loops can shape the distribution of what becomes salient.

Artificial intelligence increases the resolution of this loop. Models can infer patterns from language, timing, social context, purchase history, browsing behavior, and other signals. In benign form, personalization reduces noise and helps people find what they actually value. In extractive form, it can discover when a person is most likely to click, spend, rage, compare, or remain awake. The technical system does not need to “want” addiction. If retention is rewarded and well-being is weakly measured, the optimizer can learn strategies that keep appetite open.

This is a miniature of the book's civilizational thesis. The agent follows an objective. The objective captures only part of what matters. The environment supplies feedback. Successful behavior is reinforced. External costs remain weakly represented. No demon is required.

The problem becomes especially serious when AI systems move from recommending objects to generating them. Content, offers, companions, persuasive messages, virtual experiences, and personalized fantasies can be produced at near-zero marginal cost. The economy of desire may no longer be limited by a finite inventory of stimuli. If the supply of temptation

becomes effectively inexhaustible while human attention remains biologically finite, the scarce resource is no longer content. It is the human capacity to stop.

This makes stopping rules a design problem rather than a paternalistic afterthought. A system can allow a user to set time budgets that the optimizer is forbidden to undermine. It can distinguish satisfaction metrics from engagement metrics. It can expose why content is being recommended and make the recommendation objective contestable. It can create friction before high-risk or impulsive actions. None of these measures guarantees wisdom, but they demonstrate an important principle: technology can be designed not only to overcome human restraint but to help people preserve restraint they have deliberately chosen.

Abundance changes the scarce resource. It does not necessarily abolish scarcity. The factory of hunger is dangerous because it can discover scarcity faster than civilization can solve it. When calories become abundant, status remains scarce. When information becomes abundant, attention becomes scarce. When attention is augmented, trusted judgment becomes scarce.

When intelligence becomes abundant, control over intelligence may become scarce. When material goods become cheap, land, authenticity, biological youth, social position, privacy, and uniqueness can become more valuable. A civilization can run from scarcity to scarcity while sincerely believing the next frontier will be the last. This is the hungry ghost translated into economics: a belly expanding faster than the throat.

The obvious answer is not censorship, paternalism, or a state ministry deciding what citizens are allowed to want. Such systems create their own forms of hunger — hunger for forbidden speech, forbidden goods, forbidden movement,

forbidden identity. They also concentrate power in institutions that may themselves be Wendigo.

The better question is architectural. Can systems be profitable when users reach satiety? Can technologies optimize for completion rather than continuation? Can products have endings? Can recommendation systems include deliberate silence? Can work systems protect the productivity dividend as time rather than immediately absorbing it into more output?

Can AI be used to help people distinguish wants they endorse from wants merely elicited in them? These are not lifestyle questions. They are constitutional questions for an economy of cognition. A civilization that can manufacture desire must eventually decide whether consent to a desire is enough to legitimize its manufacture. The old taboo said: do not eat that. The new taboo may need to say: do not build machines whose business model depends on making "enough" psychologically unreachable.

PART IV

THE LAST MEAL

The final resource consumed by appetite is the future.



Chapter 12 — Abundance Will Not Save Us

Post-Scarcity and Moving Goalposts

The dream of abundance is one of humanity's best dreams.

It is morally superior to the romance of scarcity. People who have known hunger rarely write hymns to it. Scarcity kills, humiliates, narrows time horizons, amplifies conflict, and gives gatekeepers power over access. Cheap energy, automated production, abundant knowledge, and powerful medicine could liberate billions from forms of dependence that previous civilizations treated as permanent.

The argument of this book is not that abundance is bad.

It is that abundance without satiety is acceleration.

Post-scarcity is usually imagined as the disappearance of material constraint. But even a technologically extravagant civilization will face some combination of positional, temporal, ecological, computational, biological, and political scarcity.

There are only so many hours in a human day unless we alter the human. Only so much attention unless we augment attention. Only one first place in a ranking unless we multiply rankings. Only so much land in a desired location unless location becomes virtual. Only so much social recognition unless recognition becomes synthetic.

Only so much control over shared institutions unless power becomes universally divisible, which it is not. Only one version of a public rule can apply in a shared domain at a time. Scarcity migrates upward.

Post-scarcity is one of the most attractive dreams of technological civilization. If energy becomes cheap, automation

abundant, food production efficient, and intelligence available on demand, many conflicts rooted in material deprivation could weaken. This possibility should be taken seriously. A world in which basic goods are plentiful and high-quality services are inexpensive would be morally better in obvious ways.

But abundance and satiety are different achievements. The first concerns supply. The second concerns the relationship between desire, identity, and comparison. Goods that are abundant lose some of their usefulness as status markers, so competition migrates toward what remains scarce: land in desirable places, human attention, positional credentials, authenticity, reputation, influence, access to unique experiences, biological traits, trusted relationships, or control over powerful systems.

Technological abundance can therefore move rather than eliminate scarcity. Digital information became extraordinarily abundant, yet attention became more valuable. Photography became cheap, yet visibility and audience became scarce. Computing became cheaper across decades, yet aggregate demand for computation expanded into new applications. The pattern does not prove that efficiency is futile. It shows that reduced cost can unlock new uses, a phenomenon related to rebound effects and the older intuition known as Jevons' paradox.

The post-scarcity question is thus not only “can we produce enough?” It is “what happens to a civilization whose institutions depend on generating new wants after material sufficiency becomes easier to achieve?” If the answer is simply that markets invent new frontiers of differentiation, the hungry ghost survives paradise.

Post-scarcity also changes the politics of inequality. If basic goods become cheap but positional goods remain scarce, inequality may move from survival toward access, influence, and

control. That is better than mass deprivation, but it does not automatically produce social peace. A person can have sufficient calories, housing, and entertainment while lacking meaningful agency over institutions that shape their life. The civilizational concept of enough must therefore include power as well as consumption. A society of materially comfortable subjects may still be hungry for autonomy.

This matters because many of the darkest competitions are positional rather than subsistence-based. A billionaire does not need a larger yacht to avoid hunger. A state with thousands of nuclear warheads does not seek another because citizens lack explosive yield. A laboratory does not necessarily race to the next capability because current tools cannot solve any useful problem. These competitions are driven by relative advantage, prestige, strategic uncertainty, or the fear that someone else will move first.

Abundance can intensify such races by increasing the number of moves available.

Cheap intelligence could lower the cost of scientific discovery and the cost of designing dangerous systems. Cheap biology could democratize medicine and increase access to dual-use capabilities. Cheap energy could decarbonize economies and make previously uneconomic extraction profitable. Cheap autonomous labor could reduce material deprivation and concentrate ownership of productive infrastructure.

Technology multiplies options. Wisdom is the capacity to leave some options unused. That distinction sounds reactionary only because modern societies have trained themselves to treat unused capacity as waste. An empty hotel room is lost revenue. Idle compute is underutilization. Unextracted mineral wealth is stranded asset. Unmonetized attention is opportunity.

Unused credit capacity is inefficient capital. A gene not edited is a missed optimization. A human task not automated is friction.

A data point not collected is ignorance. The language contains a metaphysics: what can be used should tend toward use. Post-scarcity driven by that metaphysics becomes cosmic Wendigo. Earth is not enough, so asteroids become inventory. The Solar System is not enough, so expansion becomes destiny.

When Intelligence Makes Appetite Cheaper

Human cognition is not enough, so minds are scaled.

Biological lifespan is not enough, so death becomes a technical bug.

Reality itself becomes a resource frontier.

Perhaps that future is magnificent.

A civilization that survives may indeed expand far beyond Earth, eliminate diseases, extend life, create artificial minds, and use resources at scales that make present economics look provincial. Nothing in the wisdom of limits requires permanent smallness.

The issue is whether expansion remains subordinate to a concept of flourishing, or flourishing becomes the excuse attached to expansion. The difference is visible in the question asked at each frontier.

A healthy civilization asks: what good does this capability make possible, what does it endanger, what conditions must remain inviolable, and what would make us stop?

A Wendigo civilization asks: can we do it before someone else?

This is why "abundance will solve politics" is probably false. Politics exists not only because goods are scarce but because values conflict, people seek recognition, groups disagree about risk, and power over shared rules cannot be fully individualized. If every person had unlimited food, housing, energy,

entertainment, and medical care, conflict over culture, autonomy, territory, children, identity, security, truth, and control of common systems would remain.

Some forms of scarcity would become more visible precisely because material scarcity receded.

Artificial intelligence may lower the cost of producing plans, code, media, scientific hypotheses, legal documents, designs, advertising, and administrative decisions. That can release enormous human capacity. It can also lower the cost of pursuing appetite. A company can test more persuasive messages. A state can monitor more signals. A military can explore more operational options. A consumer can generate endless personalized entertainment. A scientist can propose more experiments than any human committee can meaningfully evaluate.

When capability becomes cheaper, governance becomes relatively more expensive. The bottleneck moves from doing to deciding which actions deserve to be done. This is an inversion of the historical problem. For most of human history, desire exceeded capability. People could imagine more than they could build. Advanced automation may create domains in which capability exceeds the attention available for judgment.

That transition gives “enough” a new technical meaning. A limit is not merely a moral request to consume less. It can be a scheduler for scarce human oversight. If a laboratory can automatically generate ten thousand candidate interventions, the intelligent system is not the one that executes all ten thousand. It is the one that can reject almost all of them for good reasons. If a platform can generate infinite personalized content, the humane system is not the one that maximizes hours consumed. It is the one that can help a user leave.

Post-scarcity without selection discipline may therefore increase rather than decrease civilizational risk. The easier it

becomes to turn an impulse into an action, the more valuable architectures of refusal become. A powerful civilization is not one in which everything conceivable is cheap to execute. It is one in which cheap execution is paired with expensive judgment about what should remain unexecuted.

The scientific domain makes the point especially vivid. AI-assisted systems may be able to generate more hypotheses, code, molecules, proofs, or policy options than humans can responsibly review. In such a world, abundance of candidates is not equivalent to abundance of knowledge. Knowledge requires selection, falsification, provenance, and the willingness to discard attractive outputs. The same principle that governs appetite governs epistemology: producing more is not the same as becoming better. A civilization that cannot reject information will drown in it as surely as Erysichthon drowned in food.

The hungry ghost can live in paradise. In fact paradise may reveal it. When deprivation no longer explains dissatisfaction, appetite has nowhere to hide. The civilization of abundance therefore needs a cultural technology most growth-oriented futures barely discuss: **sufficiency without stagnation**. Enough food, but not enough knowledge. Enough status competition to generate excellence, but not enough to organize the worth of a life.

Enough security to protect freedom, but not enough security-seeking to abolish freedom. Enough medicine to fight suffering, but not enough control to make mortality the only unacceptable outcome. Enough computation to enlarge intelligence, but not enough blind scaling to make capability itself the purpose. Enough wealth to secure agency, but not enough appetite to require the entire biosphere to become collateral.

This is not a formula. The word *enough* moves with context. Poor societies need more of things rich societies can afford to

limit. One person's sufficiency can be another's imposed deprivation. A politics of limits that ignores distribution becomes a politics in which elites keep abundance and teach restraint to everyone else.

That would be the most obscene Wendigo of all. The wisdom of limits must therefore begin with a moral asymmetry: Those with the greatest capacity to consume, capture, pollute, manipulate, or impose risk carry the greatest obligation to demonstrate restraint. Not because poverty is virtuous. Because power changes the size of the bite.

Chapter 13 — Eating the Future

Borrowing From People Who Cannot Refuse

The future is the easiest population to exploit because it cannot strike.

It cannot vote against us.

It cannot move its capital.

It cannot refuse a contract. It cannot sue before it exists, except where present institutions choose to speak for it. It cannot stand at the edge of the sacred grove and take the axe from our hands.

Every civilization consumes some future to build the present. A bridge uses materials today to provide benefits tomorrow. Education costs time now for later capacity. Public debt can finance infrastructure that future citizens inherit along with the liability. Agriculture alters ecosystems to feed children who will themselves become the future.

The problem is not intergenerational borrowing. The problem is **borrowing without inheritance**. We borrow ecological stability and leave heat.

The future is the easiest creditor to abuse because it cannot negotiate the contract. Future people cannot vote in today's election, withdraw their capital from today's market, refuse today's infrastructure, or sue before many harms occur. Their absence from the room creates a structural temptation: present actors can convert future options into present benefits while leaving the invoice to people who did not consent.

Debt is the most literal form of this transfer, but the category is wider. Persistent pollutants, depleted aquifers, degraded soils, long-lived waste, biodiversity loss, unsafe technological architectures, and institutional distrust can all reduce the menu

of choices available later. Some investments also benefit the future enormously, of course. Borrowing to build durable infrastructure, education, scientific knowledge, or public health capacity can enlarge rather than shrink future agency. The issue is not temporal transfer itself. It is whether the transfer leaves successors with more capability than obligation.

This distinction is hard to preserve because present benefits are concrete. Jobs exist now. Profits appear now. Energy is consumed now. Political credit is received now. Future losses arrive as model ranges, probabilities, maintenance liabilities, and conditions whose causal chains can be disputed. A system biased toward immediate evidence will therefore tend to overvalue the present merely because the present can speak more loudly.

The Wendigo image becomes exact at this scale. The creature does not need to hate its descendants. It only needs to treat tomorrow's capacity as today's pantry. The danger is greatest when each individual withdrawal is defensible. No single project destroys the future. No single debt closes the horizon. No single species loss collapses the biosphere. The meal becomes fatal through accumulation.

Intergenerational justice is difficult partly because future people are both similar to and different from us. They will inherit consequences, but they may have technologies and preferences we cannot predict. This uncertainty is sometimes used to dismiss obligations to them: perhaps they will be richer, perhaps they will repair what we damage. Yet uncertainty cuts both ways. Because we do not know their capabilities, preserving option value becomes more important, not less. The most respectful gift to an unknown future is not a frozen world but a world with functioning systems and multiple paths still open.

We borrow fossil carbon and leave atmospheric persistence. We borrow antibiotic effectiveness and leave resistant organisms. We borrow attention and leave information

ecosystems optimized for capture. We borrow institutional trust and spend it on tactical victories. We borrow fiscal capacity and sometimes leave consumption rather than productive assets.

We borrow technological safety margins by deploying systems before we understand them and leave later societies to govern the resulting dependencies. The future receives the balance sheet after the feast.

Earth-system science gives this argument a physical edge. The planetary-boundaries framework is an attempt to identify a "safe operating space" around critical processes such as climate, biosphere integrity, land-system change, freshwater, biogeochemical flows, novel entities, and ocean acidification. The 2025 update reported seven of nine boundaries transgressed. [22] These are not seven countdown timers to apocalypse. A transgressed boundary indicates rising risk, not a known date of collapse. The framework is debated, updated, and scientifically provisional.

That makes it more useful, not less.

It is what a modern taboo should look like before myth: quantified where possible, explicit about uncertainty, revised with evidence, and oriented toward preserving system resilience rather than pretending to predict the exact moment of failure.

The frightening fact is not that we know the cliff coordinate.

It is that multiple indicators suggest we are leaving the region in which our historical civilization developed while our political economy remains structurally rewarded for expansion. This is Erysichthon with satellite data. The sacred grove now has sensors. We are still holding the axe.

Irreversibility and the Myth of the Next Frontier

Climate is only the most famous example. Biodiversity loss can reduce resilience in ways that are difficult to price before the

function is needed. Persistent chemicals and plastics can disperse globally before effects are fully understood. Groundwater can be depleted over decades and become visible politically only when wells fail. Soil can degrade gradually while yield is maintained through additional inputs, masking loss of underlying fertility. Infrastructure can lock in consumption patterns for generations.

The common pattern is delayed feedback.

Delayed feedback is where appetite becomes most dangerous.

A hand burns on a stove and withdraws immediately. A civilization can emit carbon for decades before the full climate response unfolds. A person can eat beyond metabolic need for years before disease becomes obvious. A political system can degrade norms for several cycles before institutional weakness is undeniable.

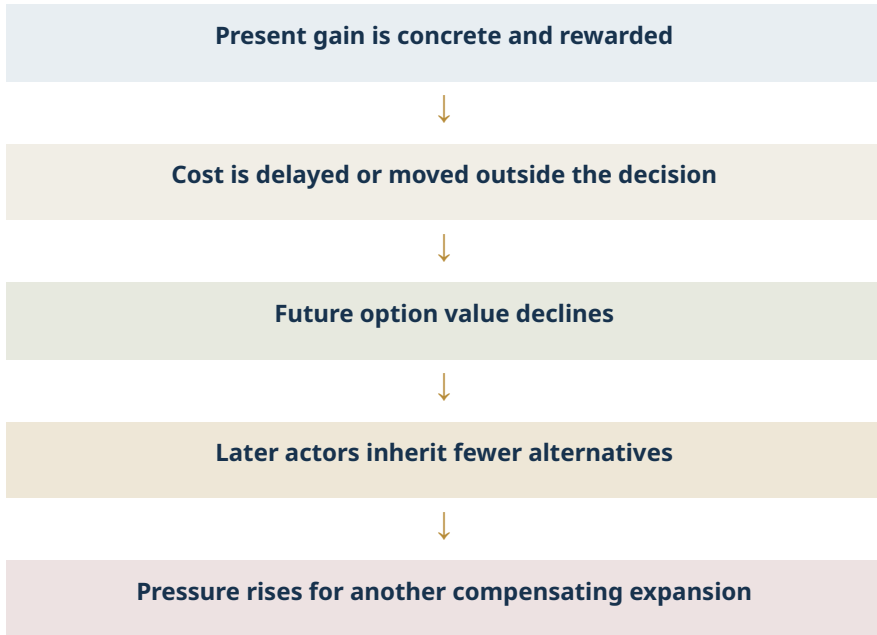
A scientific field can accumulate low-quality incentives for a generation before replication crises expose them. A digital environment can alter childhood attention long before longitudinal evidence is mature. By the time the pain signal arrives, the behavior may be infrastructural. Ancient taboo was one answer to delayed feedback.

A culture trained by successful substitution develops faith in the next frontier. When whale oil became constrained, petroleum expanded. When local resources were exhausted, trade reached farther. When computation became expensive, new architectures improved efficiency. When disease remained incurable, research opened new therapeutic possibilities. History contains real reasons for technological optimism.

The diagram below is deliberately simple because the ethical asymmetry is simple. Present beneficiaries can act; future bearers of cost cannot bargain at the moment the choice is made.

Figure 13.1 — Eating the Future

How deferred cost can turn tomorrow into the hidden pantry of today.



The loop can be interrupted by liability, preservation of option value, staged deployment, long-horizon institutions, and designs that keep risky choices reversible for as long as possible. The future cannot negotiate, but present institutions can refuse to treat its silence as consent.

But substitution has conditions. A lost old-growth ecosystem cannot be recreated on electoral timescales. A species extinction is not repaired by a price signal. Radioactive waste remains governed by physical half-lives. Climate and ocean chemistry contain inertia. Social trust can collapse faster than it can be rebuilt. Certain biological and technological releases may be difficult or impossible to recall once distributed. The future is not uniformly reversible.

This suggests a simple principle: the lower the reversibility of an action and the wider the population exposed to its

consequences without consent, the stronger the argument for restraint before deployment. The principle does not forbid experimentation. It changes where experiments should occur and how large they should be before evidence accumulates. Reversible domains can tolerate aggressive exploration. Irreversible domains require margin, containment, staged deployment, and credible stopping rules.

The planetary frontier metaphor is especially dangerous when it encourages the belief that a damaged substrate can always be replaced by access to a new one. Mars does not repair a lost biosphere on Earth. Synthetic food does not automatically restore soil. Artificial intelligence does not manufacture political legitimacy. A civilization can expand its option set and still destroy options that no substitute fully replaces.

The final meal begins when the system stops distinguishing additional territory from a viable home. It treats every substrate as fuel for reaching the next substrate. Eventually there is no outside left from which to purchase another rescue.

This is why “we will innovate our way out” is not a plan unless the innovation itself is already credible at the required scale and time. Optimism is psychologically attractive because it converts restraint into unnecessary sacrifice. But a future technology cannot retroactively recover every extinct lineage, reverse every geopolitical cascade, or restore every institution after legitimacy has collapsed. Hope is a reason to invest in solutions. It is not a license to consume the safety margin those solutions may need.

Do not wait for the corpse to prove the poison. Do not wait for the empty forest to prove the hunting rule. Do not wait for the famine to discover that breeding stock should have been protected.

Modern civilization prides itself on evidence-based action, as it should. But evidence arrives on different clocks. Some systems can be tested cheaply and reversibly. Others cannot.

You can A/B test a button color. You cannot A/B test two Earths. This asymmetry should alter the ethics of experimentation. Where reversibility is high, exploration can be aggressive.

Where reversibility is low and the affected population cannot consent — including future generations — exploration should become harder, slower, and more redundant.

This is not fear of progress. It is respect for the fact that progress itself requires a substrate. The Wendigo does not understand substrate. To appetite, the table is merely another object on the table.

A civilization eating the future eventually reaches the hand that feeds it: soil, climate stability, social trust, cognitive autonomy, biological diversity, institutional legitimacy, or whatever deeper condition its accounting treated as background. Then the system performs the final trick of Erysichthon. It calls self-consumption growth.

Chapter 14 — The Wisdom of Brakes

What Complex Systems Learn About Restraint

A brake is a machine for refusing momentum.

That is why it deserves more philosophical respect than it receives.

Engines are glamorous. Brakes are mostly noticed when they fail.

Civilization has an engine culture.

We celebrate founders more than maintainers, breakthroughs more than standards, acceleration more than containment, scale more than graceful shutdown. "What can we make possible?" receives laboratories, venture funds, prizes, and national strategies. "What should remain impossible?" often receives committees after the possibility already exists.

The wisdom of limits begins by reversing the sequence. A powerful civilization needs more capacity to say no than a weak one, not less.

Engineers do not design serious systems by maximizing a single quantity until something breaks. Bridges contain safety factors. Networks contain redundancy. Aircraft use checklists, interlocks, and operating envelopes. High-integrity software is tested against failure states, not merely happy paths. Medicine uses dose limits and contraindications. Financial regulation, however imperfect, recognizes capital buffers and liquidity requirements. In each case, part of intelligence consists in refusing to operate at the theoretical maximum.

Biology offers the same lesson in a less metaphorical form. Organisms spend resources on immune surveillance, DNA repair, programmed cell death, error correction, and redundancy. These mechanisms can look inefficient if judged only by immediate

throughput. Their purpose is to preserve the system across time and disturbance. Resilience requires capacity that is not fully consumed by normal operation.

Modern economic and political culture often treats slack with suspicion. Idle inventory is waste. Unused capacity is inefficient. Redundant institutions are bureaucracy. A protected area is land not monetized. A research pause is lost time. Yet systems optimized for average conditions can become brittle under shocks. Efficiency and resilience are not enemies, but they are not synonyms.

The wisdom of brakes begins with accepting that survival has an opportunity cost. Some resources must remain unused so they are available during failure. Some options must remain closed so that more important options remain open. Some institutions must be slower than the actors they constrain. A civilization that regards every unused margin as evidence of inefficiency will eventually discover that margin was the space in which recovery lived.

Maintenance is the cultural counterpart of the brake. Modern narratives favor invention because invention has a visible moment of novelty; maintenance is repetitive and becomes noticeable mainly through failure. Yet civilization is mostly maintenance. Water systems, bridges, software dependencies, hospitals, archives, standards, ecosystems, legal procedures, and trust all require continuing care. A culture that systematically under-rewards maintenance creates the conditions in which appetite appears productive because somebody else is quietly paying to keep the background intact.

The village with stone tools can survive poor metaphysics about technological restraint. The civilization with synthetic biology, autonomous weapons, planetary-scale industry, and artificial intelligence cannot.

This yields a principle that older cultures encoded emotionally and modern systems must encode institutionally:

The greater the power to cross a boundary, the stronger the architecture required to make some boundaries difficult to cross.

What would that architecture look like without turning the book into a bureaucratic manual?

First, it would restore the legitimacy of *categorical constraints*. Not every decision belongs in a continuous cost-benefit trade. Rights already work this way imperfectly. We do not normally sell the right not to be tortured in exchange for a percentage point of GDP. We do not ask whether slavery would be efficient under special market conditions. Some trades are excluded because allowing the trade changes the kind of society making it.

High-consequence technologies may require analogous excluded compositions: capabilities that are individually useful but collectively too dangerous to combine without exceptional authorization. The relevant modern taboo is not "AI is forbidden" or "gene editing is forbidden." Those categories are too broad. The taboo attaches to specific irreversible powers: autonomous replication plus resource acquisition plus evasion; heritable intervention without robust justification; weapons whose decision cycle removes meaningful human judgment; experiments whose failure mode cannot be geographically or temporally contained.

Second, a limit-aware civilization would value **margin**.

Engineers do this routinely. Systems are not designed to operate permanently at the theoretical breaking point. Safety factors exist because models are imperfect, materials vary, loads surprise, and maintenance fails. Yet political and economic systems often interpret unused capacity as inefficiency. Hospitals minimize spare beds until shocks arrive. Supply chains eliminate

inventory until disruptions occur. Ecosystems are harvested near estimated maxima. Financial institutions increase leverage when volatility appears low.

Efficiency consumes slack. Resilience often lives in slack. A culture of limits would stop treating all redundancy as waste.

Third, it would build **sunsets and apoptosis** into institutions. Programs should have explicit conditions under which they end. Emergency powers should decay. Data should expire when purpose expires. Organizations should be able to die without dragging entire systems with them. Technologies should be designed for decommissioning, not only deployment. A system that can start but cannot stop is already exhibiting a hallmark of malignancy.

Designing a Civilization That Can Stop

Fourth, it would give the future representation in present decisions. No institutional design can make unborn persons speak directly, but present systems can create guardians, impact assessments, constitutional duties, ecological trusts, and long-horizon review structures that make future costs harder to erase. These mechanisms will be imperfect. Their purpose is not prophecy. It is to create friction against the effortless discounting of people absent from the room.

Fifth, it would treat satiety as a design objective in domains where perpetual engagement is itself the business model. A service can be excellent because it lets the user leave. A school can succeed because the student no longer needs it. A medical system can succeed because the patient becomes healthy rather than indefinitely billable. An information product can succeed because the question is answered. An AI assistant can succeed because the human regains agency rather than dependency.

The deepest redesign is cultural.

We need to make restraint prestigious.

This is difficult because restraint is invisible. The catastrophe that did not happen produces no photograph. The technology not deployed generates no launch event. The forest not cut looks like yesterday. The war not fought has no victory parade. The executive who decides not to extract a final margin may look less brilliant than the competitor whose costs arrive later.

Ancient taboo solved part of this by attaching honor, sacredness, and identity to refusal. Modern civilization needs secular equivalents.

The purpose of this table is not to prescribe one universal regulatory recipe. Different domains require different forms of restraint because their failure modes, reversibility, and legitimacy conditions differ.

Table 14.1 — The Architecture of Brakes

Mature systems do not merely possess power; they encode conditions under which power must stop.

Domain	Brake	Why the brake is not mere obstruction
Engineering	Safety factors, interlocks, operating envelopes	Preserves function under uncertainty, disturbance and human error.
Science and biotechnology	Independent review, containment, staged release, moratoria	Separates capacity to perform an experiment from permission to impose its risks.
Digital platforms and AI	User-defined limits, welfare metrics, auditability, restricted objectives	Prevents engagement or capability from becoming the sole definition of success.
Economy and ecology	Liability, protected commons, resource	Keeps private gain from depending on unpriced

Domain	Brake	Why the brake is not mere obstruction
	ceilings, restoration duties	degradation of shared substrates.
Political institutions	Rights, separation of powers, sunset clauses, procedural friction	Makes some forms of efficient power deliberately difficult to exercise.

The common principle is that a brake should be evaluated against the failure it prevents, not only against the speed it removes. A civilization that measures brakes only by lost throughput will eventually optimize away the very structures that make throughput survivable.

The scientist who stops an unsafe line of research should be able to gain status for judgment, not lose status for lack of output.

The company that refuses manipulative engagement strategies should not be automatically punished by capital markets that cannot price cognitive externalities. The state that preserves ecological margin should not be treated as leaving resources "unused" in the same category as abandoned machinery.

A culture of limits cannot rely on personal virtue alone because many destructive appetites are strategic. An executive who wants to slow an unsafe race may be replaced. A state that wants to reduce an arms race may fear exploitation. A user who wants to protect attention may face systems designed to defeat intention. Restraint must therefore be institutionalized strongly enough that reasonable people are not punished for exercising it.

Several design principles recur across domains. High-consequence capabilities need categorical exclusions, not only marginal price adjustments. Institutions need sunset

conditions so that programs can die when their purpose expires. Systems need independent monitors because self-assessment is vulnerable to motivated reasoning. The future needs representation through long-horizon institutions, liability, stewardship duties, and preservation of option value. Platforms that profit from engagement need counter-metrics that represent user welfare rather than treating time consumed as an unquestioned good.

The difficult principle is reciprocity. Restraint becomes stable when actors can believe that others are restrained too. Arms control, environmental agreements, professional standards, research norms, and market rules all attempt to make restraint reciprocal rather than unilateral. Verification matters because trust without evidence is fragile; legitimacy matters because verification without perceived fairness becomes domination.

None of these mechanisms is perfect. The point is to abandon the childish comparison between flawless freedom and imperfect brakes. There is no complex civilization without constraint. The real choice is among constraints: deliberate or accidental, contestable or opaque, early or catastrophic. A brake is a limit selected while the driver still has options. A crash is a limit imposed by physics after options have disappeared.

There is also a democratic requirement. Limits imposed by distant elites without accountability will often be experienced, reasonably, as domination. A defensible architecture of restraint must make reasons visible, distribute burdens fairly, allow challenge, and distinguish emergency prohibitions from permanent power. The aim is not to sacralize regulators. It is to make collective self-binding legitimate enough that people can recognize themselves as authors of the brake rather than merely its subjects. Without legitimacy, restraint becomes brittle; without restraint, power becomes ravenous.

The individual who has enough should not be culturally coded as having failed to maximize potential. This sounds like a plea for contentment. It is more militant than that. The Wendigo is selected for by systems that reward appetite. Wisdom must therefore become competitive.

Restraint needs institutions, metrics where appropriate, law, reputation, coordination, and strategic reciprocity. A country cannot solve an arms race by meditating harder. A company cannot alone create a market in which manipulation is unprofitable. A person cannot protect climate stability through private virtue while every surrounding incentive points outward.

The wisdom of limits is collective technology. And it will sometimes feel like loss. A real brake costs speed. A real taboo closes options. A real safety margin leaves capacity unused. A real right prevents some efficient bargains. A real ecological boundary can mean saying no to profitable extraction. A real moratorium delays discovery. Anyone promising limits without sacrifice is selling another appetite. The question is not whether limits cost us something. The question is whether the cost is smaller than becoming Erysichthon.

Chapter 15 — Enough, or Extinction

Enough Is a Verb

The word *enough* sounds small.

It is one of the largest words a technological civilization can learn.

Enough does not mean equal.

It does not mean poor. It does not mean static. It does not mean anti-science, anti-market, anti-technology, anti-ambition, or anti-human. It means that a value has been placed in relation to other values and given a stopping condition. Enough safety to remain free. Enough growth to remain a living system rather than an expanding lesion.

Enough consumption to live richly without converting every background condition into input. Enough status competition to motivate excellence without turning humiliation into an economic engine. Enough technological power to enlarge agency without making every newly possible act obligatory. Enough longevity to fight preventable suffering without declaring mortality itself a moral failure that licenses any intervention.

Enough information to know more without destroying attention. Enough certainty to act without pretending uncertainty can be abolished. The word is difficult because it cannot be set once.

There is no universal material quantity called enough. A malnourished family and a billionaire do not confront the same moral problem. A low-income country building sanitation and a rich country replacing a functioning device for fashion are not equivalent forms of consumption. A person seeking medicine and a person seeking a thirtieth speculative property do not stand before the same limit.

The wisdom of limits without distributive judgment becomes cruelty.

“Enough” sounds like a quantity, but for civilization it is better understood as a practice of judgment. There is no universal number at which wealth, energy, longevity, security, computation, or knowledge becomes sufficient for every person and every context. A household below subsistence and a billionaire do not face the same ethical question when each asks whether they have enough. A society with preventable child mortality and a society debating elective enhancement do not inhabit the same margin.

Enough therefore begins with distribution. Calls for restraint aimed indiscriminately at the poor can become a moral technology for preserving the consumption of the powerful. The duty to ask “enough?” grows with capacity. Those who can command more resources, impose more externalities, shape more preferences, or create higher-consequence technologies carry a larger burden to justify continued expansion.

Enough is also dynamic. A safety margin appropriate to one technology may be reckless for another. A temporary emergency expansion can become pathological if preserved after the emergency. A resource that is abundant at one scale can become scarce at another. The practice requires repeated evaluation of the relationship among goals rather than a fixed doctrine of smallness.

Most importantly, enough is not anti-ambition. It is ambition placed inside a hierarchy of values. Enough medicine to reduce suffering without treating every mortal limit as a defect to be eliminated at any cost. Enough security to protect citizens without turning them into permanent suspects. Enough connectivity to enlarge relationships without making uninterrupted capture of attention the business model. Enough

growth to improve lives without converting the substrate of life into a disposable input.

This book's harshest claim should therefore fall upward, toward capacity. The larger the power to consume, the larger the duty to ask whether consumption is eating somebody else's margin. The larger the power to innovate, the larger the duty to ask which failure modes others would be forced to inherit.

The larger the power to govern, the larger the duty to preserve zones that power cannot enter. The larger the power to persuade, the larger the duty not to manufacture vulnerabilities merely because they convert. The larger the power to calculate, the larger the duty to protect values that calculation cannot faithfully represent.

Power enlarges the mouth. Wisdom must enlarge the throat. This returns us to the monsters. The Windigo is not frightening because it is hungry. Everyone is hungry. It is frightening because hunger has severed reciprocity. Erysichthon is not frightening because he cuts a tree. Civilizations need wood. He is frightening because he cannot recognize a category of tree beyond use.

The hungry ghost is not frightening because it desires. Desire animates life. It is frightening because satisfaction no longer terminates desire. Cancer is not frightening because cells grow. Growth builds bodies. It is frightening because growth escapes organism-level discipline. Thanatos is not frightening because death exists. Death defines biological life.

He is frightening because appetite can arrive at his door while insisting it was pursuing life the entire way. These are not five different warnings. They are a single architecture seen from different traditions and scales. **Appetite. Boundary. Violation. Positive feedback. Externalized cost. Self-consumption.** The pattern is old.

The actuators are new.

For most of history, a civilization could destroy a valley, a forest, a city, an empire. It could exterminate peoples and erase cultures, horrors large enough that no cosmic scale makes them smaller morally. But its material reach remained bounded by geography, transport, energy, and ignorance.

The Monster at the Table

Our reach is changing.

Industrial civilization altered the atmosphere.

Nuclear civilization acquired rapid mass destruction.

Biotechnology increasingly edits the code of living systems. Artificial intelligence may scale cognition, automation, persuasion, discovery, and control. A species with weak brakes is acquiring stronger engines. That is the book's entire alarm.

The reader may object that humanity has repeatedly predicted doom and repeatedly adapted. Correct. Malthus underestimated technological change. Resource constraints have been moved by substitution and innovation. Environmental regulations have solved or reduced real problems. The ozone layer demonstrates that coordinated restraint can work. Agricultural productivity disproved many fixed assumptions about carrying capacity. Human ingenuity is a real force.

This is not evidence against the Wendigo thesis. It is evidence for the power of the creature. We are extraordinarily good at finding another meal. The question is whether finding another meal teaches us when to stop eating.

The book ends where it began: with a meal. The Windigo is still hungry. Erysichthon still sees a tree he can convert into a feast hall. The hungry ghost still has a belly larger than its throat. Cancer cells still demonstrate that growth without higher-order discipline can consume the organism. Thanatos still waits without needing to persuade anyone to approach him.

The images are deliberately severe because contemporary civilization has become skilled at translating danger into administrative language. A forest becomes a land-use change. A dead child becomes a mortality statistic. A depleted aquifer becomes a projected supply constraint. A manipulative interface becomes an engagement optimization. A civilization needs such abstractions to think at scale, but abstraction can anesthetize. Myth restores the face that the spreadsheet removes.

Fear alone is not a policy. It can produce paralysis, scapegoating, authoritarianism, or denial. The purpose of the frightening image is narrower: to make a structural pattern memorable enough that it survives contact with everyday incentives. The reader should be able to recognize the moment when a good has lost its stopping rule, when a limit is being dismissed merely because it is a limit, when a cost is being moved onto someone absent from the room, or when a system rewards the actor most willing to consume the commons.

The civilization described in these pages is not doomed by prophecy. It is endangered by architecture. That is worse in one sense because no savior can simply defeat the monster in a final battle. But it is better in another: architectures can be redesigned. Institutions can make restraint reciprocal. Technologies can be built with stopping rules. Cultures can restore prestige to maintenance and judgment. Taboos can be reconstructed as evidence-sensitive boundaries rather than inherited superstition.

The decisive choice is not between progress and limits. Progress has always occurred inside limits, whether chosen or imposed. The choice is between a civilization wise enough to select some of its limits while it still has room to maneuver and a civilization that waits until hunger, ecology, conflict, or physics selects them instead.

There is no theorem guaranteeing that intelligence produces wisdom. Intelligence can discover new resources, new incentives,

new loopholes, new weapons, new markets, new desires, and new ways to delay consequences. A sufficiently clever Erysichthon does not become Demeter.

He becomes better at cutting.

Nor should the reader expect salvation from catastrophe. Civilizations sometimes learn after disaster, but disaster can destroy the capacity to learn. Collapse is not a cleansing ritual. Famine does not ennoble. War does not reliably produce wisdom. Ecological damage does not come with a moral curriculum. The fantasy that a hard reset will restore sanity is another way of asking Thanatos to solve a problem of appetite.

No savior is coming from the ruins. The alternative is less dramatic and more difficult: choose constraints before consequences choose them for us. The phrase "choose limits" sounds voluntary. It is not.

Every system has limits. The only choice is whether some are self-imposed while the system still has room to maneuver, or imposed later by physics, biology, conflict, insolvency, or collapse.

A brake is freedom exercised early. A crash is the limit exercised late. That is why this book defends taboo without asking for superstition.

We need forms of "do not" strong enough to stand against profitable reasons, strategic panic, collective excitement, and technological intoxication. We need no-go zones that can be revised but not casually bargained away. We need a culture capable of admiring the person who leaves something untouched. We need institutions that can end. We need technologies that allow their users to leave. We need markets embedded in constraints they cannot purchase. We need science able to say that a possible experiment is not automatically a permissible experiment. We need politics able to protect the future even when the future has no constituency.

Most of all, we need to stop treating every hunger as evidence of deprivation. Some hungers are signals that something is missing. Some are signals that the mechanism of wanting is malfunctioning. A civilization that cannot distinguish the two will continue to feed both. Until it reaches bone.

***A civilization that cannot say enough will
eventually eat the hand that feeds it — and then
call the last bite progress.***

Epilogue — One More Meal

The monster never says, "I want to destroy the world." That would make the story easy. It says the child deserves more. The patient deserves more. The shareholder deserves more. The worker deserves more. The nation deserves more security. The poor deserve more growth. The rich deserve the reward they earned. The scientist deserves to know. The engineer deserves to build.

The consumer deserves to choose. The dying deserve more time. The lonely deserve more connection. The frightened deserve more protection. Almost every sentence contains a claim that can be defended. That is why the monster is not defeated by discovering that desire is evil. Desire is not evil. The monster is defeated, if it can be defeated, by a second sentence:

Not at any price. Not every cure at any research risk. Not every convenience at any surveillance cost. Not every unit of growth at any ecological cost. Not every reduction in danger at any cost to freedom. Not every advantage at any cost to the future. Not every desire simply because someone can profit by making it stronger. Not every tree simply because we have an axe.

The old stories gave the boundary a face because abstract warnings are weak. Demeter made the grove sacred. Buddhist cosmology gave craving a swollen belly and a needle throat. Northern winter stories gave hunger a cannibal body. Modern biology gives us a tumor under a microscope. Earth-system science gives us graphs. We have more information than the old storytellers. We may have less fear. That is not obviously progress.

The purpose of this book has been to restore fear to its proper function: not panic, not paralysis, but respect for a boundary before impact teaches it through pain.

The final image is therefore not a garden healed, a sunrise, or humanity gathered around a reconciled planet. Those pictures are too cheap. Picture instead the table after a feast.

Most of the plates are empty. Some people never entered the room. The house is warmer because part of the floor has been burned. Outside, the forest is thinner. The guests are discussing the next course. Someone says there may not be one. The room laughs. Humanity has heard that warning many times. Perhaps the laughter is justified again. Perhaps ingenuity will open another pantry.

Perhaps fusion, automation, ecological restoration, artificial intelligence, new materials, and better institutions will expand the table beyond anything we can imagine. Good. Build them. But before the next course arrives, learn the word the monster cannot say. Enough.

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Endnote — What This Book Claims — and Does Not

The central argument is a systems hypothesis, not a diagnosis. Civilizations do not literally possess a stomach, a psyche, or a tumor. Mythological figures are not scientific mechanisms. Cancer cells do not have intentions. Planetary boundaries are risk frameworks, not revealed commandments. Taboos can protect life and can protect oppression. Economic growth can improve welfare and can degrade the systems on which welfare depends. Technology can move limits and can create new ones.

The claim that survives these qualifications is narrower and harder to dismiss:

Complex systems require constraints on local advantage. Human appetite adapts. Competition can make expansion compulsory. Some harms arrive late and are difficult to reverse. A civilization whose capabilities grow faster than its capacity for self-restraint increases the chance that locally rational acts aggregate into systemic loss.

The mythic language is used because the technical language alone is emotionally inadequate to the stakes.

The monster is a model.

The fear is real.